

Machine Learning in Applications

AM/04 - Dataset Augmentation for Additive Manufacturing defect detection

Giorgia Modi, Giacomo Ponzuoli, Erika Genova, Marco Ammirati
Politecnico di Torino

CONTENTS

I	Introduction	1
II	Background	1
III	Materials and methods	2
III-A	Dataset	2
III-B	Implementation details	2
III-B1	Variational Autoencoder (VAE)	2
III-B2	Conditional VAE (CVAE)	3
III-B3	Generative Adversarial Network (GAN)	3
III-B4	Single-Image GAN (SinGAN)	3
III-B5	Diffusion Models	3
III-C	Metrics	3
IV	Results and discussion	4
IV-A	Classifier on original images	4
IV-B	Variational Autoencoders (VAEs)	4
IV-C	Conditional Variational Autoencoders (CVAEs)	5
IV-D	Deep Convolutional Generative Adversarial Networks (DCGANs)	6
IV-E	Single Image Generative Adversarial Networks (SinGANs)	7
IV-F	Stable Diffusion	8
V	Conclusions and future works	9
	References	10
	Appendix A: VAEs Experimental Configurations	11
	Appendix B: VAEs Qualitative Results	11
	Appendix C: CVAEs Experimental Configurations	15
	Appendix D: CVAEs Qualitative Results	15
	Appendix E: GANs Experimental Configurations	19
	Appendix F: GAN Qualitative Results	19
	Appendix G: SinGAN Experimental Configurations	24
	Appendix H: SinGAN Qualitative Results	24
	Appendix I: StableDiffusion Experimental Configurations	28
	Appendix J: StableDiffusion Qualitative Results	29

LIST OF FIGURES

1	Powder Bed Fusion (PBF) process	1
2	Example images from original dataset.	2
3	Representative examples of the best results obtained using the VAE model with the Fusion_7_18 and Fusion_11_22 configurations for the <i>Defects</i> class.	5
4	Representative examples of the best results obtained using the CVAE model with the Exp9 configuration for the <i>Defects</i> class.	6
5	Representative examples of the best results obtained using the DCGAN model with the Fusion_9_16 and Fusion_12_13 configurations for the <i>Defects</i> class.	7
6	Representative examples of the best results obtained using the SinGAN model with the Comb_Exp4 configuration for the <i>Defects</i> class.	8
7	Examples illustrating the impact of different parameter settings in image generation using Stable Diffusion model.	10
8	Qualitative results of Fusion_1_12. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	11
9	Qualitative results of Fusion_2_13. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	12
10	Qualitative results of Fusion_3_14. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	12
11	Qualitative results of Fusion_4_15. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	12
12	Qualitative results of Fusion_5_16. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	12
13	Qualitative results of Fusion_6_17. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	13
14	Qualitative results of Fusion_7_18. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	13
15	Qualitative results of Fusion_8_19. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	13
16	Qualitative results of Fusion_9_20. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	13
17	Qualitative results of Fusion_10_21. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	14
18	Qualitative results of Fusion_11_22. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	14
19	Qualitative results of Exp1. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	15
20	Qualitative results of Exp2. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	15
21	Qualitative results of Exp3. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	16
22	Qualitative results of Exp4. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	16
23	Qualitative results of Exp5. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	16
24	Qualitative results of Exp6. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	16
25	Qualitative results of Exp7. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	17
26	Qualitative results of Exp8. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	17
27	Qualitative results of Exp9. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	17
28	Qualitative results of Exp10. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	17
29	Qualitative results of Exp11. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	18
30	Qualitative results of Exp1 (<i>Defects</i>)	19
31	Qualitative results of Exp2 (<i>Defects</i>)	19

32	Qualitative results of Exp3 (<i>Defects</i>)	20
33	Qualitative results of Exp4 (<i>Defects</i>)	20
34	Qualitative results of Exp5 (<i>Defects</i>)	20
35	Qualitative results of Exp6 (<i>NoDefects</i>)	20
36	Qualitative results of Exp7 (<i>Defects</i>)	21
37	Qualitative results of Exp8 (<i>Defects</i>)	21
38	Qualitative results of Exp9 (<i>NoDefects</i>)	21
39	Qualitative results of Exp12 (<i>Defects</i>)	21
40	Qualitative results of Exp13 (<i>NoDefects</i>)	22
41	Qualitative results of Exp14 (<i>NoDefects</i>)	22
42	Qualitative results of Exp15 (<i>Defects</i>)	22
43	Qualitative results of Exp16 (<i>Defects</i>)	22
44	Qualitative results of Exp17 (<i>Defects</i>)	23
45	Qualitative results of Exp18 (<i>Defects</i>)	23
46	Qualitative results of Exp1 (<i>Defects</i>)	24
47	Qualitative results of Exp2 (<i>Defects</i>)	24
48	Qualitative results of Exp3 (<i>Defects</i>)	25
49	Qualitative results of Exp4 (<i>Defects</i>)	25
50	Qualitative results of Exp5 (<i>Defects</i>)	25
51	Qualitative results of Exp6 (<i>Defects</i>)	25
52	Qualitative results of Exp7 (<i>Defects</i>)	25
53	Qualitative results of Exp8 (<i>Defects</i>)	26
54	Qualitative results of Exp9 (<i>Defects</i>)	26
55	Qualitative results of Exp10 (<i>Defects</i>)	26
56	Qualitative results of Exp11 (<i>Defects</i>)	26
57	Qualitative results of Exp12 (<i>Defects</i>)	26
58	Original Images. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	27
59	Comb_Exp1. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	27
60	Comb_Exp2. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	27
61	Comb_Exp3. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	27
62	Comb_Exp4. The first two images belong to the <i>Defects</i> class, while the last two belong to the <i>NoDefects</i> class.	27
63	Qualitative results of Exp1 (<i>Defects</i>)	29
64	Qualitative results of Exp2 (<i>Defects</i>)	29
65	Qualitative results of Exp3 (<i>Defects</i>)	29
66	Qualitative results of Exp4 (<i>Defects</i>)	29
67	Qualitative results of Exp5 (<i>Defects</i>)	30
68	Qualitative results of Exp6 (<i>Defects</i>)	30
69	Qualitative results of Exp7 (<i>Defects</i>)	30
70	Qualitative results of Exp8 (<i>Defects</i>)	30
71	Qualitative results of Exp9 (<i>Defects</i>)	30
72	Qualitative results of Exp10 (<i>Defects</i>)	31
73	Qualitative results of Exp11 (<i>Defects</i>)	31
74	Qualitative results of Exp12 (<i>Defects</i>)	31
75	Qualitative results of Exp13 (<i>Defects</i>)	31
76	Qualitative results of Exp14 (<i>Defects</i>)	31

LIST OF TABLES

I	Quantitative results of VAE experiments. Metrics include LPIPS (lower is better) and classification accuracy on GEN_train and GEN_test for models trained from scratch or with pretrained initialization.	4
II	Quantitative results of CVAE experiments.	5
III	Quantitative results of GAN experiments.	6
IV	Quantitative results of combined SinGAN-based experiments.	7
V	Quantitative results of StableDiffusion experiments.	9
VI	Summary of experimental configurations for VAE training runs.	11
VII	Summary of experimental configurations for CVAE training runs.	15
VIII	Summary of experimental configurations for DCGAN training runs.	19
IX	Configuration parameters for single SinGAN experiments.	24
X	Configuration parameters for combined SinGAN experiments.	24

XI	Configuration parameters for StableDiffusion LoRA experiments. Prompt and Negative Prompt refer to the textual variants listed in Table XII.	28
XII	Prompt variants (A–E) used across experiments.	28
XIII	Negative prompts (F–G) used across experiments.	28

Machine Learning in Applications

AM/04 - Dataset Augmentation for Additive Manufacturing defect detection

Abstract—Additive Manufacturing (AM), particularly through Powder Bed Fusion (PBF), enables the production of complex components but still presents challenges in quality control. Machine Learning algorithms can help automate defect detection; however, the limited availability of defective samples compromises model performance. This work explores several synthetic image generation techniques to face this problem: Variational Autoencoders (VAE), Conditional VAE (CVAE), Deep Convolutional GAN (DCGAN), SinGAN, and Stable Diffusion. Each model was evaluated using quantitative metrics (LPIPS, GEN_train, GEN_test) and visual inspection. Results show that while VAE and CVAE produce structurally coherent images, they often lack fine detail. DCGAN and SinGAN achieve more realistic results but exhibit higher variability. Stable Diffusion consistently delivers high visual quality and generalization, emerging as the most effective approach for dataset augmentation in industrial scenarios. The synthetic images generated in this study lay the groundwork for future work on more complex tasks such as defect localization and multi-class classification.

I. INTRODUCTION

Metal Additive Manufacturing (AM) is a cornerstone of Industry 4.0, offering the ability to produce complex but high-quality parts directly from a Computer-Aided Design (CAD) model using a layer-by-layer construction method [1].

One of the most prominent techniques within metal AM is Powder Bed Fusion (PBF), a process in which a thin layer of metal powder is spread over a build platform and selectively fused using a high-energy heat source—such as a laser or electron beam—according to the CAD data (Figure 1). This process is repeated layer by layer, enabling the creation of complex geometries with excellent mechanical properties and minimal material waste [2] [1].

Despite its potential, PBF still faces several quality monitoring challenges that limit its adoption in large-scale production. In particular, quality control remains a labor-intensive and error-prone task, often requiring human inspection and costly testing procedures. In this context, Computer Vision and Machine Learning (ML) algorithms can support the automation of quality control tasks.

However, training accurate classifiers to distinguish between defective and non-defective parts requires a large and balanced dataset—something often unavailable in real production lines. In particular, generating a sufficient number of images showing actual defects is costly and time-consuming due to the expensive nature of the AM process and the difficulty of reproducing specific anomalies on demand.

To address this data limitation, recent research has turned to generative models, which have shown significant promise in

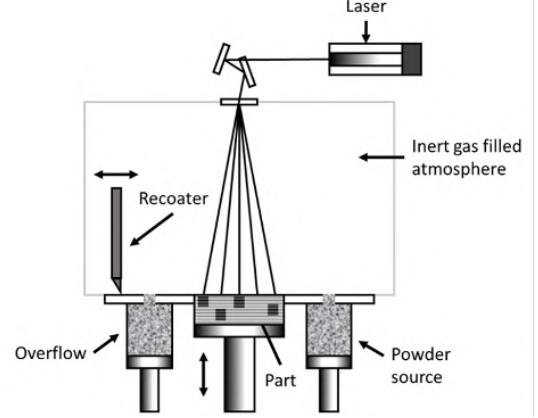


Figure 1: Powder Bed Fusion (PBF) process

synthesizing realistic data across various domains. By generating synthetic images, these models enable the augmentation of limited datasets, improving class balance and enhancing the generalization ability of machine learning classifiers.

In this work, generative models are employed to create realistic images of both defective and non-defective parts, reducing reliance on physical sample production and demonstrating their effectiveness in improving classification performance for PBF quality inspection.

This paper is organized as follows. Section II introduces the generative techniques employed in this study. Section III describes the dataset, implementation details, and evaluation metrics. Section IV presents the experimental findings based on the application of different generative approaches to synthesize images for dataset augmentation in the context of defect classification. Finally, Section V concludes the paper and discusses directions for future research.

II. BACKGROUND

As outlined in Section I, quality control in PBF can benefit from data-driven approaches powered by synthetic data generation. This section presents the generative models used for data augmentation in this study.

Conventional data augmentation techniques—such as rotation, scaling, or translation—are often insufficient for addressing class imbalance and introducing meaningful variability. Instead, this work leverages Generative Data Augmentation (GDA) [3], where new synthetic images are created using deep generative models, enhancing both the quantity and diversity of the training dataset.

A range of architectures has thus been investigated, spanning from early Variational Autoencoders (VAEs) [4] and their conditional variants (CVAEs) [5], to Generative Adversarial Networks (GANs) [6]. For scenarios with extremely limited data, Single Image GANs (SinGANs) [7] have been utilized to learn multi-scale textures from single images, while Diffusion Models [8] have been applied to achieve superior fidelity and diversity in image synthesis.

Variational Autoencoders (VAEs) encode each input into a continuous latent distribution, from which new images are sampled. While this probabilistic framework is efficient, the use of pixel-wise losses (e.g., L1 or L2) smooths high-frequency details, often yielding perceptually blurry outputs. Conditional VAEs (CVAEs) incorporate class labels into both encoder and decoder, enabling targeted and class-specific synthesis.

Generative Adversarial Networks (GANs) replace reconstruction losses with an adversarial game between generator and discriminator, producing sharper, more detailed images than VAEs. However, GANs demand careful hyperparameter tuning, can exhibit training instability, and occasionally collapse to limited output modes, thereby reducing diversity. In this work, Deep Convolutional GANs (DCGANs) [9] are adopted, which stabilize training through the use of convolutional and transposed convolutional layers, batch normalization, and appropriate activation functions. This architecture is well-suited for generating synthetic defect and non-defect images to augment the dataset.

SinGAN learns from a single image by building a multi-scale pyramid of generator-discriminator pairs trained sequentially from coarse to fine resolutions. This enables the model to capture both global structure and local textures without requiring large datasets, making it ideal for scenarios with extremely limited data.

Diffusion models achieve state-of-the-art image quality by first corrupting images with noise through a forward diffusion process and then learning a parametric reverse process that denoises step by step. This two-stage framework avoids the blurriness of VAEs and the mode-collapse issues of GANs, delivering high fidelity and diverse samples. In particular, this work adopts Stable Diffusion, a type of Latent Diffusion Model (LDM) [10], which applies the diffusion process in a learned, lower-dimensional latent space obtained via an autoencoder, rather than directly in high-dimensional pixel space. This approach drastically reduces computational cost and memory requirements—by orders of magnitude—while preserving high visual fidelity. As a result, it is particularly well-suited for high-resolution image synthesis, domain-specific fine-tuning, and text-to-image generation tasks. However, diffusion models, including LDMs, face certain limitations. A notable challenge is the poor time efficiency during inference. The iterative denoising steps required for image generation are computationally intensive, leading to longer inference times compared to other generative models.

A recent comparative analysis of generative techniques confirms these trade-offs, highlighting how VAEs, GANs, and Diffusion Models each offer distinct advantages depending on task requirements and constraints [11].

III. MATERIALS AND METHODS

This section outlines the implementation details of the generative modeling approaches employed to synthesize both defective and non-defective images, as well as the characteristics of the dataset and the evaluation metrics used to assess image quality and downstream classification performance.

A. Dataset

This project utilizes a private dataset of grayscale images acquired from a Powder Bed Fusion Additive Manufacturing process. These images capture individual layers of metallic powder during the printing operation and have been provided specifically for this study. The dataset is organized into two distinct classes:

- **Defects:** it contains 47 images of different layers with one or multiple defects in each of them without labeling.
- **NoDefects:** it contains 33 images of different layers without defects.

Representative examples from the original dataset are shown in Figure 2.

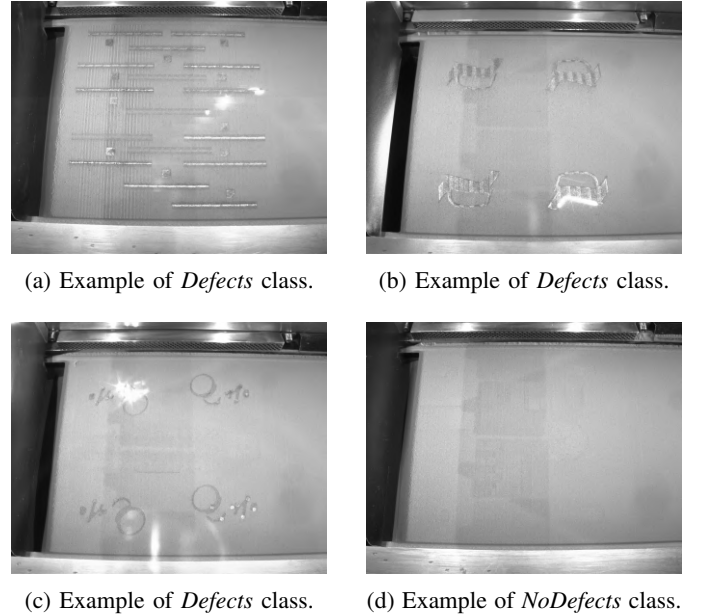


Figure 2: Example images from original dataset.

B. Implementation details

All generative models were implemented in PyTorch and executed on NVIDIA GPUs. Model-specific hyperparameters (e.g. image size, latent dimensionality, number of scales) were supplied at runtime via command-line flags and changed based on the experiment; architectures dynamically adapt to these values.

1) *Variational Autoencoder (VAE)*: A convolutional VAE was implemented to accept user-specified image size H and latent dimension L via command-line arguments. The architecture comprises:

- **Encoder:** four 5×5 convolutional layers with strides of 2 and paddings of 2, producing feature maps of sizes

$\{16, 32, 64, 128\}$. The spatial size after each layer is computed at runtime by

$$s_{i+1} = \lfloor (s_i - k + 2p)/2 \rfloor + 1,$$

where $k = 5$, $p = 2$. The final feature map is flattened to size $128 \times s_4^2$, projected through a 512-unit dense layer, and then split into mean μ and log-variance $\log \sigma^2 \in \mathbb{R}^L$.

- **Reparameterization:** latent vectors z sampled as $z = \mu + \epsilon \cdot \exp(0.5 \log \sigma^2)$ with $\epsilon \sim \mathcal{N}(0, I)$.
- **Decoder:** mirrors the encoder in reverse: a linear layer expands z back to the flattened feature size, then four transposed convolutions (5×5 , stride = 2, padding = 2, output padding = 1) to reconstruct a $1 \times H \times H$ image, ending in a sigmoid activation.
- **Loss & Optimization:** the reconstruction loss is the sum of per-pixel MSE and the Kullback–Leibler divergence weighted by a factor w_{KLD} (default = 1). Model parameters were optimized using the Adam optimizer with default learning rate 10^{-3} and weight decay 10^{-3} , both configurable via command-line flags.

2) *Conditional VAE (CVAE)*: The CVAE extends the VAE by conditioning on a one-hot class vector (*Defects* vs. *NoDefects*) of length 2:

- **Encoder:** concatenates the class map to the image tensor, applies two 5×5 convolutions (stride = 2), flattens to 300 units, then computes $\mu, \log \sigma^2 \in \mathbb{R}^L$.
- **Decoder:** concatenates the sampled latent vector with the class vector, then uses two linear layers and three transposed convolutions (5×5 , 5×5 , 4×4) to reconstruct the image.
- **Loss & Optimization:** same as the VAE.

3) *Generative Adversarial Network (GAN)*: As previously discussed, a Deep Convolutional GAN (DCGAN) architecture was adopted due to its proven stability and performance in image synthesis tasks. The image size H and latent dimension L are configurable via command-line arguments. The architecture is as follows:

- **Generator:** maps $z \in \mathbb{R}^L$ through a linear layer into a $128 \times (H/4) \times (H/4)$ tensor, then applies two up-sampling blocks (each: Upsample $\rightarrow$ Conv2d(3 $\times$ 3) $\rightarrow$ BatchNorm2d $\rightarrow$ LeakyReLU) and a final Conv2d(3 $\times$ 3) $\rightarrow$ tanh to produce the output image.
- **Discriminator:** four downsampling blocks (each: Conv2d(3 $\times$ 3, stride = 2) $\rightarrow$ LeakyReLU $\rightarrow$ Dropout; with BatchNorm2d except in the first), yielding feature maps $\{16, 32, 64, 128\}$. The final features are flattened and passed to a linear layer followed by a sigmoid activation for real/fake classification.
- **Regularization:** optional R1 gradient penalty [12] on real samples, weighted by λ (default 10).
- **Loss & Optimization:** binary cross-entropy adversarial loss; Adam optimizers with separate learning rates (η_G, η_D) and betas (β_1, β_2) all configurable at runtime.

4) *Single-Image GAN (SinGAN)*: The original SinGAN framework was employed to learn a multi-scale pyramid of GANs from each single defect image. The official public

repository¹ was cloned and modified with minor adjustments to support our data loader and grayscale normalization. The codebase was further simplified retaining only the training routines necessary for this project.

5) *Diffusion Models*: Diffusion-based augmentation was performed using the Stable Diffusion 1.5 Img2Img pipeline² from HuggingFace’s Diffusers library³. Initially, the pre-trained pipeline was applied directly to defect and non-defect images, with strength, guidance scale, and inference steps varied via command-line arguments. Subsequently, the pipeline was fine-tuned on the metal AM dataset using DreamBooth LoRA—a DreamBooth-based [13] low-rank adaptation method [14] for rapid fine-tuning—where instance and class prompts distinguished defective from smooth powder-bed scenes. Fine-tuning was conducted at a resolution of 512 px with a learning rate of 10^{-5} . In addition to these base settings, other hyperparameters were selectively modified across different experiments to evaluate their influence on generation quality and training stability. At inference, the resulting LoRA weights were loaded to generate defect images under the same adjustable CLI parameters.

C. Metrics

To assess the quality of the synthetic images and their usefulness in improving classification, three evaluation metrics were employed:

- **LPIPS (Learned Perceptual Image Patch Similarity)** [15]: a perceptual similarity metric that measures how visually similar two images are. In this project, LPIPS has been adapted: each synthetic image has been compared to all real images from its corresponding class, and the minimum LPIPS score was taken. The final LPIPS value reported is the average of these minimum scores across all generated samples.
- **GEN-Train**: a classifier is trained on real images and tested on synthetic ones. It measures the realism and diversity of the generated data. Higher test accuracy values indicate that the generated images are close enough to the real ones to be correctly recognized by a model that was never exposed to them during training.
- **GEN-Test**: a classifier is trained on synthetic images and tested on real ones. It evaluates the usefulness of synthetic images for training. Higher values indicate that the synthetic data captures relevant features, making them effective for training models.

GEN-Train and GEN-Test metrics were computed using a convolutional neural network classifier based on the ResNet18 architecture. Both a model trained from scratch and one initialized with ImageNet-pretrained weights were used. For all the experiments the model was trained using the Adam optimizer with a learning rate of 10^{-3} , over 50 epochs and a batch size of 4.

In addition to quantitative metrics, all generated images were also evaluated through visual inspection. This qualitative

¹SinGAN GitHub repository

²Stable Diffusion Img2Img API

³Diffusers GitHub repository

assessment focused on texture quality and the realism of the synthesized defects. In fact, quantitatively evaluating image generation experiments remains a challenging task, as standard metrics often fail to fully capture perceptual quality. Moreover, many commonly used metrics—such as FID (Fréchet Inception Distance) and IS (Inception Score)—require a large number of samples to be computed reliably. Due to the limited size of our dataset, these metrics were not suitable for a meaningful evaluation. As a result, visual inspection remains the most reliable and useful way to judge how realistic and accurate the generated images are in our case.

IV. RESULTS AND DISCUSSION

This section presents the experiments conducted on both defect and no-defect image synthesis. For each experiment, a total of 80 images were generated, consisting of 40 defective and 40 non-defective samples. Configuration details and a large set of qualitative results are provided in the appendix for each experiment, divided by generative model, offering additional insight into the outcomes achieved (see Appendix A and B for VAE, Appendix C and D for CVAE, Appendix E and F for GAN, and Appendix I and J for StableDiffusion).

A. Classifier on original images

As a baseline experiment, a ResNet classifier was trained on the original dataset to distinguish defective from non-defective parts. Different architectures such as ResNet18, ResNet34, and ResNet50 were considered, allowing flexibility in the choice of backbone. Training employed a standard k-fold cross-validation scheme with a held-out test split containing unseen data. Both pretrained weights (ImageNet initialization) and training from scratch were evaluated. Subsequently, simple data augmentations including RandomHorizontalFlip, RandomVerticalFlip and RandomAdjustSharpness were applied during training. Training parameters are the same as those used for the GEN-Train and GEN-Test metrics.

The classifier rapidly converged, reaching 100% accuracy on both training and validation sets within a few epochs. Test accuracy also reached 100% for both pretrained and scratch-trained models. The augmented training yielded the same near-perfect accuracy results as training on the original dataset alone. The exceptional performance is likely due to the clear visual distinction between defective and non-defective samples, combined with low intra-class variability. Notably, nearly all non-defect images (except for six) correspond to empty layers, while these six images are more challenging to distinguish from defect images. Given the dataset’s characteristics, the binary classification task is relatively straightforward for the model to learn, enabling perfect accuracy on unseen data.

Consequently, the focus of this work is not on improving the classifier performance, which is already near-perfect, but rather on generating synthetic defect images to enable more complex downstream tasks such as defect localization and classification of multiple defect types.

Exp	LPIPS ↓	GEN_train ↑		GEN_test ↑	
		Scratch	Pretrained	Scratch	Pretrained
Fusion_1_12	0.3285	1.0000	1.0000	1.0000	0.4750
Fusion_2_13	0.3325	0.5000	1.0000	1.0000	0.5000
Fusion_3_14	0.2961	1.0000	1.0000	1.0000	0.9125
Fusion_4_15	0.3164	0.6125	1.0000	1.0000	0.9125
Fusion_5_16	0.3297	0.9000	1.0000	1.0000	1.0000
Fusion_6_17	0.3779	1.0000	1.0000	0.5250	0.4125
Fusion_7_18	0.2940	1.0000	1.0000	1.0000	0.9750
Fusion_8_19	0.3159	0.5125	1.0000	0.6125	0.7750
Fusion_9_20	0.2712	1.0000	1.0000	0.8625	0.3375
Fusion_10_21	0.3277	1.0000	1.0000	1.0000	1.0000
Fusion_11_22	0.2878	0.7000	1.0000	1.0000	1.0000

Table I: Quantitative results of VAE experiments. Metrics include LPIPS (lower is better) and classification accuracy on GEN_train and GEN_test for models trained from scratch or with pretrained initialization.

B. Variational Autoencoders (VAEs)

This section presents the results obtained using a VAE for image generation. To avoid manual labeling of the synthesized data, each experiment was performed twice—once on defect images and once on non-defect images—using identical hyperparameters. The results from the two runs were then combined into a single fusion, resulting in 80 generated images per experiment. For instance, Fusion_1_12 denotes the combination of Exp1 (defect) and Exp12 (non-defect). In total, 11 such fusion experiments were carried out, as shown in Table I, varying five key hyperparameters: number of training epochs, latent space dimensionality, image resolution, learning rate, and the weight of the Kullback-Leibler (KLD) term. The batch size was fixed at 4 in all experiments.

The initial focus was placed on the latent dimension. In Fusion_1_12, a latent vector of size 128 was used, with image dimensions of 512×512 px and 100 training epochs. In this configuration, the *Defects* class produced visually poor outputs, where defects were barely visible or entirely absent; images generated for the *NoDefects* class exhibited more defined powder bed contours, even though still affected by a general blurriness (see Figure 8). Regarding quantitative metrics, Fusion_1_12 shows a slightly high LPIPS value, indicating limited perceptual similarity with real samples. However, it achieves excellent GEN_train and GEN_test classification scores. This result may be attributed to the classifier’s ability to accurately recognize the more realistic and consistent images generated for the *NoDefects* class.

In Fusion_2_13, only the latent dimension was increased to 256; this change led to further degradation of the *Defects* images, which featured dark patches and visual artifacts, as can be seen in Figure 9. *NoDefects* images remained relatively clean. This is also reflected in a slight worsening of the quantitative metrics.

For Fusion_3_14, training was extended to 500 epochs, with the latent dimension still set to 256. This was the first experiment to produce clearly visible, well-defined, and diverse defects in the *Defects* class. The generated defects appeared in various forms and locations, making them more representative and realistic (see Figure 10). The *NoDefects* class, instead, benefited from slightly enhanced sharpness. Consistent with

the visual improvements, the quantitative metrics also show positive results, with a notable improvement in the LPIPS score.

In *Fusion_7_18* experiment, the number of training epochs was further increased to 600 compared to *Fusion_3_14*, resulting in additional, though modest, improvements in both visual quality and quantitative metrics (Figure 14). These results suggest that the latent dimension and the number of training epochs significantly influence image quality, particularly for the *Defects* class, which appears more sensitive to these hyperparameters.

Starting from the baseline experiment *Fusion_1_12*, where the learning rate was set to 10^{-3} , the learning rate was lowered to 10^{-4} in *Fusion_4_15*. This adjustment did not lead to substantial improvements in overall image quality; however, the defects became visually more pronounced, as shown in Figure 11.

In *Fusion_6_17*, the image resolution was increased to 800×800 pixels. While this led to a slight improvement in image definition, it did not result in noticeable enhancements in overall image quality or defect visibility (Figure 13).

Another series of experiments explored the effect of reducing the KLD weight (originally set to 1). A lower KLD penalty typically improves reconstruction quality by allowing the decoder to make greater use of the latent code. However, this can lead to a less structured latent space and a mismatch between training and inference: during generation, sampling from the prior may yield latent codes the decoder has not seen, resulting in poor outputs. In *Fusion_8_19*, with the KLD weight set to 0.5 and 200 epochs, *Defects* images appeared blurry and lacked meaningful defects (Figure 15). In *Fusion_9_20*, increasing training to 500 epochs led to slight improvements in both sharpness and defect presence, although diversity remained limited (Figure 16). Further increasing the number of epochs to 700 in *Fusion_11_22* made the defects more visible (Figure 18), resulting in the best-performing experiment in this group.

In relation to the observations made above, the best experiments in terms of visual quality were *Fusion_7_18* and *Fusion_11_22*, with representative examples shown in Figure 3.

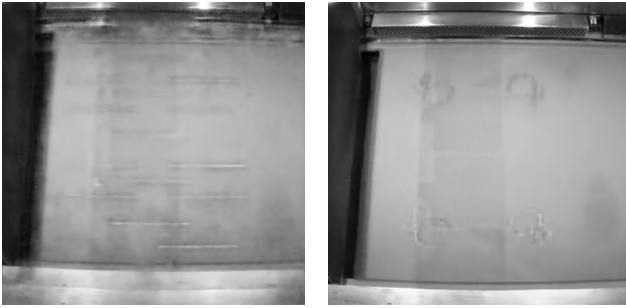


Figure 3: Representative examples of the best results obtained using the VAE model with the *Fusion_7_18* and *Fusion_11_22* configurations for the *Defects* class.

In the end, the **VAE model** is capable of generating visually diverse defects, with variation across experiments. In some

cases, multiple defects were observed within the same image, suggesting that the model successfully learns a sufficiently complex distribution. Despite this, as expected, the VAE-generated images tend to be somewhat blurry and not entirely realistic. Meanwhile, synthetic images from the *NoDefects* class remained nearly identical across all experiments—likely due to the low variability in the original dataset.

Indeed, in accuracy tests conducted on *GEN_train* using a *pretrained* initialization, all experiments achieved perfect accuracy (1.0). Moreover, in six out of eleven experiments, even the *scratch* models reached the same result. This behavior can be attributed to systematic issues in *Defects* images—blurriness, darkness, staining—that make them easily distinguishable from the more regular and uniform *NoDefects* images. Such a strong visual difference likely simplifies the classification task, enabling models to identify defect-free images easily and assign all others to the opposite class, thereby achieving high accuracy.

C. Conditional Variational Autoencoders (CVAEs)

Exp	LPIPS ↓	GEN_train ↑		GEN_test ↑	
		Scratch	Pretrained	Scratch	Pretrained
Exp1	0.2533	0.5625	0.5625	1.0000	0.9250
Exp5	0.3456	0.6000	0.5500	1.0000	1.0000
Exp6	0.5509	0.4250	0.7000	0.8875	0.9625
Exp7	0.3257	0.5625	1.0000	0.5875	0.5875
Exp8	0.6724	0.3375	0.7750	0.4750	0.9125
Exp9	0.3277	1.0000	1.0000	1.0000	0.8500
Exp10	0.3689	0.9000	0.6875	1.0000	0.5875
Exp11	0.2643	0.8750	1.0000	1.0000	0.6000

Table II: Quantitative results of CVAE experiments.

This section presents the results obtained using a CVAE for image generation. A total of 11 experiments were conducted, displayed in Table II. For some experiments, such as those from Exp2 to Exp4, quantitative metrics were not computed because the models performed very poorly; as a result, these experiments are not included in the table.

Unlike the VAEs, where each experiment required two separate trainings (one for *Defects* and one for *NoDefects* images), the CVAEs incorporate class labels directly into the model. This allows training a single model per experiment on both classes simultaneously, significantly reducing computational time. The CVAE experiments closely follow the logic and setup of those performed for the VAEs, varying the same five hyperparameters.

Exp1 produced the lowest LPIPS score among all CVAE and VAE experiments, indicating high perceptual similarity to original images. However, defect visibility was limited, as can be seen from Figure 19. This unusually low LPIPS can be explained by the metric implementation (see Section III-C), as some original defect images contain nearly empty layers where defects are barely perceptible, causing the metric to favor these as “similar” images.

Starting from the settings of Exp1, increasing the number of training epochs in Exp5 improved visual quality, with defects becoming more distinguishable (Figure 23). Exp2,

which increased the latent dimension to 256, performed very poorly, generating mostly noise (Figure 20). This issue was partially resolved by extending training to 600 epochs in Exp7, which yielded substantially improved image quality, although defects remained visually barely noticeable (Figure 25). Notably, defect and no-defect images generated by Exp7 appeared almost indistinguishable, reflected in very low GEN_train and GEN_test scores due to the classifier’s difficulty in differentiating the classes.

Exp4 investigated the effect of lowering the learning rate to 10^{-4} , resulting in significantly degraded outputs that appeared gray and lacked clear features (Figure 22).

Increasing image resolution to 800×800 px in Exp6 led to poorer results, with images becoming darker, more blurred, and occasionally containing incoherent artifacts (see Figure 24), as also reflected in inferior quantitative metrics.

Regarding the KLD weight, Exp8 through Exp11 showed trends similar to those observed in the VAE experiments. Halving the KLD weight to 0.5 and training for 200 epochs (Exp8) produced grainy and poorly defined images (Figure 26). Increasing epochs to 500 (Exp9) and 700 (Exp11) substantially improved image quality (Figure 27 and Figure 29).

It is worth noting that the CVAE model employed fewer layers than the VAE (two layers compared to four layers), which may explain the overall lower quality of the generated images.

However, among the experiments, Exp9 yielded the best synthetic images, as shown qualitatively and quantitatively. Figure 4 presents examples of these images, illustrating the CVAE’s performance under optimal training.

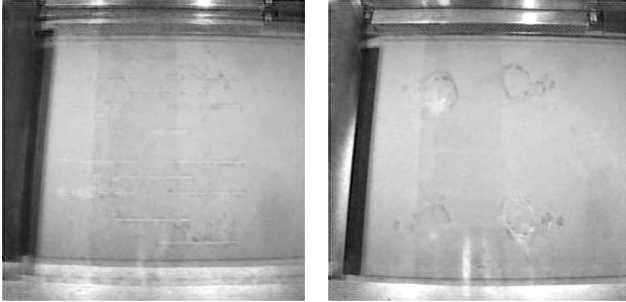


Figure 4: Representative examples of the best results obtained using the CVAE model with the Exp9 configuration for the Defects class.

Overall, while the CVAE demonstrates promise in generating images conditioned on defect labels, the use of a simplified architecture in the present implementation, combined with the limited ability to synthesize diverse defect combinations, underscores the need for further refinement. Future work may investigate architectural enhancements or alternative training strategies to more effectively capture the complex distributions and interactions of multiple defect types.

D. Deep Convolutional Generative Adversarial Networks (DCGANs)

Exp	LPIPS ↓	GEN_train ↑		GEN_test ↑	
		Scratch	Pretrained	Scratch	Pretrained
Fusion_1_6	0.4621	1.0000	1.0000	1.0000	0.4250
Fusion_2_14	0.6500	0.5000	0.5000	0.5375	0.5375
Fusion_9_8	0.4707	0.9667	0.8000	0.8500	0.6875
Fusion_9_16	0.4516	1.0000	1.0000	0.8750	0.6875
Fusion_13_12	0.5372	0.9875	1.0000	0.6000	0.6500
Fusion_13_17	0.4890	1.0000	0.8875	0.6500	0.5750
Fusion_13_18	0.5350	0.8375	1.0000	0.6125	0.5750

Table III: Quantitative results of GAN experiments.

A series of DCGAN experiments were conducted to synthesize defect and non-defect images under varying training conditions. Following the same procedure used for the VAE experiments, each DCGAN setup was run twice—once on defect images and once on non-defect images—using identical hyperparameters. The two runs were then fused into a single result comprising 80 synthesized images. The experiments were carried out by varying several hyperparameters, including the number of epochs, latent space dimensionality, output resolution, generator and discriminator learning rates (lr_G , lr_D), and the use of R1 regularization. The quantitative results are summarized in Table III.

The first test (Fusion_1_6) used a 512×512 px image resolution, a latent dimension of 128, and 1500 training epochs. Visual inspection of Exp1 (Defects images) showed that the model generated realistic outputs in which defects were clearly visible (Figure 30). The corresponding non-defect run (Exp6) produced clean and coherent textures, free of artifacts (Figure 35). The combined result, Fusion_1_6, achieved perfect GEN_train accuracy (1.0000), indicating that a latent size of 128 is sufficient to generate high-quality 512×512 px images. In contrast, increasing the latent dimension to 256 (Exp3), led to over-parameterization and lower visual quality in the outputs (Figure 32).

In Fusion_2_14, the image size was increased to 800×800 px while keeping the latent dimension at 128. This led to higher-quality and more defined outputs in the defect images from Exp2 compared to those from Exp1 (Figure 31). However, defect patterns were still not clearly represented. In particular, the non-defect images from Exp14 appeared very similar to the defect ones (Figure 41), suggesting limited variability. This is reflected in low GEN_train/GEN_test accuracy (around 0.50), indicating that the classifier struggled to distinguish between the two classes. Moreover, the LPIPS score was particularly high (0.6500), highlighting a lack of perceptual diversity in the generated images despite their apparent visual sharpness.

In Exp7, R1 regularization was added to the training setup used in Exp2. This modification resulted in only marginal improvements compared to the non-regularized version, as can be seen in Figure 36, suggesting that the original configuration in Exp2 was already stable enough without additional regularization.

Experiments 4 and 5 involved modifying the generator and discriminator learning rates for the synthesis of defect images.

These changes caused mode collapse or excessive noise (see Figure 33 and Figure 34), so the default learning rate of 0.0002×10^{-4} was maintained for all subsequent experiments.

Exp8 (*Defects*) and Exp9 (*NoDefects*) were conducted at 1024×1024 px resolution with a latent size of 128 (*Fusion_9_8* in Table III). The images generated in Exp8 exhibited enhanced detail, with most defect features clearly visible and well-preserved (see Figure 37), and those from Exp9 were highly defined and realistic (see Figure 38).

In Exp16 (*Defects*), R1 regularization was introduced using the same configuration as Exp8, resulting in *Fusion_9_16* (as the *NoDefects* experiment (Exp9) remained unchanged). This modification led to a further reduction in LPIPS to 0.4516 and achieved perfect GEN_train accuracy along with strong GEN_test performance. Visual inspection confirms that the generated images maintain a consistently high level of quality (see Figure 43), although they appear slightly more blurred compared to those from Exp8.

Directly increasing the latent dimension to 256 at 1024×1024 px resolution proved infeasible due to memory limitations. As a compromise, Exp12 (*Defects*) and Exp13 (*NoDefects*) were conducted at 800×800 px with a latent size of 256, resulting in *Fusion_12_13*. The generated images appeared darker but introduced novel and well-defined defect patterns (see Figure 39). Notably, a distinctive arrangement of circular spots, resembling a grid of small "dots", emerged in several synthesized samples. This pattern, which was present in some original training images, had not been successfully reproduced by previous models. Its appearance here suggests an improved ability of the generator to capture and replicate more complex defect structures. This improvement is likely due to the increased latent dimensionality, which provided a richer input space for the generator to map from, enabling finer and more detailed image synthesis.

Starting from the Exp12 configuration, doubling the number of training epochs in Exp17 (*Defects*) while keeping the same settings (800 px resolution, latent size 256) led to a visible improvement in image quality. The generated samples appeared more coherent and realistic (see Figure 44), and the LPIPS—evaluated via *Fusion_13_17* (with Exp13 as the unchanged *NoDefects* component)—slightly decreased to 0.4890. Despite these visual improvements, GEN_test accuracy remained low, indicating that the synthetic data still struggled to generalize to real distributions.

In Exp18, R1 regularization was added to the same setup used in Exp17. This change had a clear visual impact: the generated images appeared more similar to those from earlier experiments, with brighter textures and a return of previously observed patterns (see Figure 45). However, this came at the cost of slightly worse quantitative results. In this case, R1 regularization did not provide clear benefits and may have counteracted some of the improvements achieved by extended training.

For all the experiments, as expected, GEN_test remained lower than GEN_train across all fusions, highlighting the challenge of training models using only synthetic data. Interestingly, in some configurations, models trained from scratch outperformed their pretrained counterparts—especially

on GEN_test. This may be due to a mismatch between the features learned during pretraining and the specific distribution of the target task. Pretrained models can carry inductive biases that are suboptimal for detecting fine artifacts in generated images, leading to under-adaptation during fine-tuning and poorer generalization. In contrast, training from scratch enables the model to tailor its representations more effectively to the characteristics of synthetic data, thus improving classification performance under certain conditions.

Overall, the experiments with GANs, while producing realistic results in several cases, did not yield outstanding quantitative performance. This may be due to the fact that, while some generated images appear visually convincing, there is considerable inconsistency among images produced in the same experiment—many being low quality and noisy. Such variability is a common consequence of the inherently unstable training process of GANs and negatively impacts the evaluation metrics. Nevertheless, the best trade-offs were achieved by *Fusion_9_16* and *Fusion_12_13*, and some of the best results are shown in Figure 5.

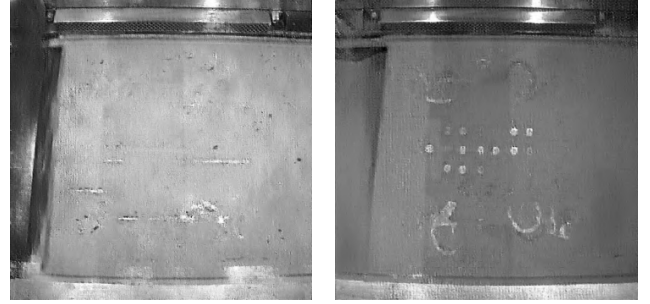


Figure 5: Representative examples of the best results obtained using the DCGAN model with the *Fusion_9_16* and *Fusion_12_13* configurations for the *Defects* class.

E. Single Image Generative Adversarial Networks (SinGANs)

Comb_Exp	LPIPS ↓	GEN_train ↑		GEN_test ↑	
		Scratch	Pretrained	Scratch	Pretrained
Comb_Exp1	0.3111	1.0000	1.0000	0.9875	0.7500
Comb_Exp2	0.3451	0.7500	1.0000	0.6000	0.4125
Comb_Exp3	0.3150	1.0000	1.0000	0.9875	0.8875
Comb_Exp4	0.3054	1.0000	1.0000	0.8875	0.7500

Table IV: Quantitative results of combined SinGAN-based experiments.

This section reports the results of experiments performed with SinGAN models. The study focused on the effects of variations in key training parameters, including the scale_factor, which determines the scaling between successive image pyramid levels; n_iter, the number of training iterations per scale; the learning rates lr_g and lr_d for the generator and discriminator networks, respectively; and the number of update steps for each network at every scale, denoted as G_steps and D_steps.

The SinGAN training process was divided into two phases. In the first phase, given that SinGAN operates on a single image, one with rich detail from the *Defects* class was

selected to enable the generation of higher-quality outputs while varying hyperparameter settings. In the second phase, the chosen configuration was applied to a larger number of images—including both *Defects* and *NoDefects* classes—to generate synthetic images for subsequent quantitative evaluation.

Numerous experiments were conducted during the initial phase of hyperparameter tuning, but the three described below were the most significant.

In Exp1 (scale factor = 0.50, 150 iterations, $lr_g = 0.001$, $lr_d = 0.0005$, 7 generator updates, 5 discriminator updates), the outputs exhibited texture discontinuities and blurred details, due to rapid scale transitions and insufficient convergence (see Figure 46). In fact, when the scale factor is too low, the transition between scales — i.e., the increase in image resolution at each scale — becomes too rapid, resulting in poor visual quality.

In Exp4, increasing the scale factor to 0.80 and the number of iterations to 600 helped the model to combine information more effectively between scales, resulting in a more coherent global structure. However, fine details remained largely absent (see Figure 49).

In Exp11, all parameters from Exp4 were maintained, except for an increased scale factor of 0.90, a reduced discriminator learning rate of $lr_d = 0.0001$, and an increased number of training iterations to 2000. This adjustment was made to prevent the discriminator from overpowering the generator and to enhance the model’s ability to recognize finer details within the image. The synthetic images in this configuration showed the best visual quality among the three, featuring more consistent detail and fewer artifacts. This suggests that a higher scale factor and a relatively weaker discriminator can lead to more effective refinement of texture details (see Figure 56).

This initial phase yielded four well-performing parameter configurations, reported in Table X. These configurations were used in the second phase to perform multiple identical training runs (in terms of parameters) on different images—specifically, two images from the *Defects* class and two from the *NoDefects* class (the same images used across all configurations). This approach was adopted to introduce greater variety into the synthetic dataset, which would otherwise have consisted of highly similar images if generated from a single input image. To maintain consistency with all experiments involving generative models, 40 synthetic images were generated per class, resulting in a total of 80 synthetic images, which were used to compute quantitative evaluation metrics.

Table IV reports the quantitative metrics obtained from the four experiments. Figure 59, 60, 61, 62 reports examples of generated images, along with the original images (Figure 58).

In Comb_Exp2, the discriminator’s learning rate was reduced relative to Comb_Exp1. Consequently, the LPIPS metric increased while the GEN_test score decreased significantly, indicating a deterioration in performance. This outcome aligns with visual inspection, as Comb_Exp2 produced the poorest results both compared with Comb_Exp1 and with all other experiments.

Raising the scale factor from 0.80 to 0.90 (Comb_Exp2 vs. Comb_Exp3) led to improvements in both LPIPS and test accuracy, suggesting that a finer pyramid scaling better preserves structural detail. This effect is also visible in the generated images—particularly in the first two images from the left, which exhibit defects—where Comb_Exp3 clearly reproduces considerably more detail.

Finally, substantially reducing the generator and discriminator learning rates (Comb_Exp3 vs. Comb_Exp4) yielded the lowest LPIPS score (0.3054), indicating enhanced perceptual similarity to real images. Experiment Comb_Exp4 emerged as the best of the four, both visually and according to the evaluation metrics. Figure 6 presents the images generated in this experiment, which represent the highest-quality results for the single-image GAN.

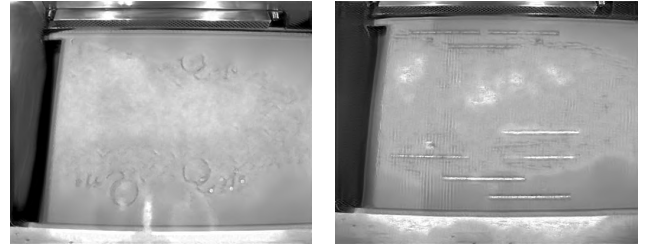


Figure 6: Representative examples of the best results obtained using the SinGAN model with the Comb_Exp4 configuration for the *Defects* class.

Overall, the results show that the generated images achieve low LPIPS values, with classifier generalization depending on training strategy: training on real data and testing on synthetic data yields reliable accuracy, while training on synthetic data requires careful model initialization to perform well on real images. However, qualitative inspection revealed a persistent limitation—distortion or noise consistently affects the central region of synthesized images. This issue likely originates from the input image characteristics and the SinGAN architecture, which, originally designed for natural images with rich textures and color variation, struggles to model grayscale technical images with reduced structural complexity and limited texture, limiting its ability to generate spatially coherent outputs in critical areas.

F. Stable Diffusion

A series of experiments were conducted using diffusion models to evaluate their capacity to generate high-quality synthetic images of powder bed layers. Specifically, as explained in Section III-B5, Stable Diffusion 1.5 Img2Img pipeline was used in these experiments. This pipeline takes an input image and a text prompt, and generates a new image by gradually refining the input to align with the semantic content described in the prompt, enabling controlled image-to-image translation. Table V summarizes the quantitative results obtained.

Several key parameters were varied across the experiments to evaluate their impact on generation quality. These can be grouped into two categories:

Exp	LPIPS ↓	GEN_train ↑		GEN_test ↑	
		Scratch	Pretrained	Scratch	Pretrained
Exp1	0.2369	1.0000	0.8333	1.0000	1.0000
Exp2	0.2384	1.0000	1.0000	1.0000	1.0000
Exp3	0.2359	1.0000	1.0000	1.0000	1.0000
Exp4	0.1988	1.0000	1.0000	1.0000	1.0000
Exp5	0.4531	0.9881	0.9167	1.0000	1.0000
Exp7	0.1945	1.0000	1.0000	1.0000	1.0000
Exp8	0.2177	1.0000	1.0000	1.0000	1.0000
Exp9	0.2229	1.0000	1.0000	1.0000	1.0000
Exp10	0.2295	1.0000	1.0000	1.0000	1.0000
Exp11	0.2508	1.0000	1.0000	1.0000	1.0000
Exp12	0.2099	1.0000	1.0000	1.0000	0.9875
Exp13	0.2155	1.0000	1.0000	1.0000	1.0000
Exp14	0.2380	1.0000	1.0000	1.0000	1.0000

Table V: Quantitative results of StableDiffusion experiments.

1) Generation parameters (Stable Diffusion Img2Img):

- strength: controls how much the original image is altered. Lower values preserve more of the input structure, while higher values allow for greater transformation.
- guidance_scale: determines how strongly the model follows the text prompt during generation. Higher values produce outputs that are more aligned with the prompt.
- Prompt: the textual input provided to guide image generation.
- Negative prompt: specifies undesirable features to avoid during generation.

2) Fine-tuning parameters (DreamBooth LoRA):

- rank: defines the dimensionality of the low-rank adaptation matrices in LoRA. Lower ranks reduce model complexity and memory usage.
- lora_dropout: introduces dropout in the LoRA layers to improve generalization, especially useful in low-data regimes.
- train_text_encoder: if enabled, allows the text encoder to adapt during fine-tuning, improving alignment with domain-specific vocabulary.
- steps: Specifies the number of fine-tuning steps used in training the LoRA-adapted model.

The experiments went through different phases: in the early phase (Exp4 and Exp5), no fine-tuning was performed and only the strength and guidance_scale parameters were varied, using a general prompt. These preliminary tests revealed the importance of the strength parameter, as lower strength values better preserved the original content, producing images that maintained the visual characteristics of the defects (Exp4, Figure 66), while higher values led to unrealistic or too noisy outputs (Exp5, Figure 67), as illustrated in the Figure 7b.

Subsequently, in Exp1–2–3, the model was fine-tuned on the AM dataset. These experiments focused on evaluating the effects of different values for guidance_scale and rank, while still relying on a general-purpose prompt. Both qualitatively and quantitatively, these experiments yielded excellent results: the defects were clearly highlighted, and the original image was only slightly altered, preserving the overall structure and appearance (see Figures 63, 64, 65).

Once a baseline fine-tuned model configuration was established, subsequent experiments focused on improving prompt precision and the training setup. In particular, the prompts were rewritten to include more details about defect characteristics, while also clarifying the appearance of non-defective surfaces (Exp6–7–8, Figures 68, 69, 70). In addition, from experiment 8, training settings were expanded: longer training schedules were adopted, introducing `train_text_encoder`, to allow better adaptation to technical terminology, and `lora_dropout`, to improve regularization.

From these experiments emerged a key strength of diffusion models: their dual ability to either faithfully replicate the input image or apply controlled modifications, for example, slightly shifting the position of a defect while preserving the overall structure. An example of this can be seen in Figure 70, where small square-shaped defects are reproduced in different locations across otherwise similar images. This behavior is particularly valuable for data augmentation, as it enables the generation of highly realistic variations of existing samples, thereby enriching the dataset with meaningful diversity without compromising visual fidelity.

The following set of experiments (from Exp9 to Exp14) mainly explored the impact of the guidance_scale parameter, which was varied between 9 and 12 to observe how much the prompt influenced the generation. Based on earlier findings, strength was kept low (typically 0.15–0.25), as this had proven to produce the most structurally coherent results (see Figures 71, 72, 73, 74, 75, 76).

From a visual point of view, when the prompts explicitly required the creation of new defects in different positions and the guidance scale was set to high values, the model occasionally introduced unwanted noise. This was particularly noticeable in the form of excessive hole-like defects (Figure 7c). However, this visual noise did not significantly affect the quantitative results: LPIPS scores remained low, and both GEN_train and GEN_test classifiers were consistently able to distinguish between classes, probably because the added noise does not compromise the overall structure or shape of the defects. However, pushing this behavior to the extreme may risk producing images that no longer appear realistic.

Overall, all diffusion-based experiments yielded strong results, both in terms of perceptual similarity and classification accuracy. These outcomes were in line with expectations, since diffusion models represent the state of the art in the image generation tasks. The models produced visually convincing outputs across a wide range of parameter configurations, confirming their robustness even when trained on limited industrial datasets. This highlights the suitability of diffusion approaches for augmenting datasets in real-world scenarios.

V. CONCLUSIONS AND FUTURE WORKS

This work presented a comprehensive comparison of generative models for dataset augmentation in the context of Additive Manufacturing (AM), specifically focusing on the Powder Bed Fusion (PBF) process. Five different approaches were explored — Variational Autoencoders (VAE), Conditional VAE (CVAE), Deep Convolutional GANs (DCGAN), SinGAN, and

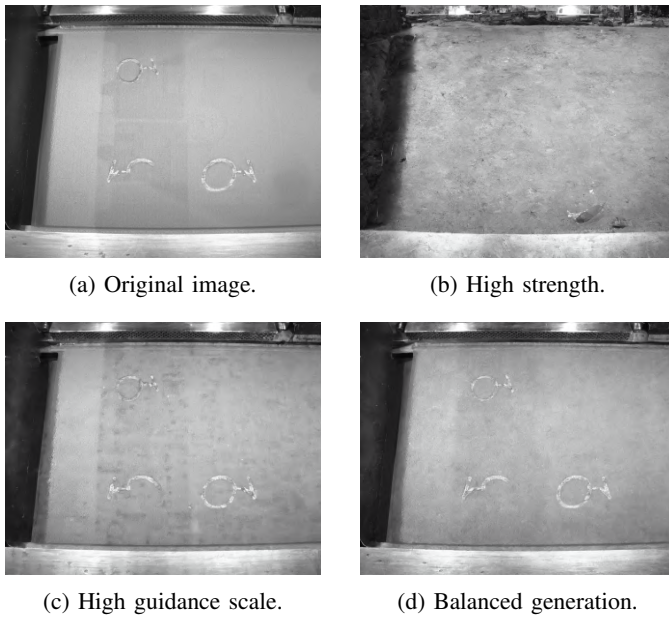


Figure 7: Examples illustrating the impact of different parameter settings in image generation using Stable Diffusion model.

Stable Diffusion— with the final goal of generating realistic images of defective and non-defective powder bed layers.

VAE and CVAE models showed potential in capturing the general structure of the data but suffered from blurry outputs and lack of fine detail, particularly for the *Defects* class. Increasing training epochs and reducing the KLD weight improved the image quality slightly, though performance remained limited compared to other methods.

DCGANs generated more detailed and varied images, especially when trained at higher resolutions and with appropriate regularization. Increasing the latent space dimensionality also led to improved results, as it allowed the generator to encode and reproduce more complex visual patterns. However, training remained highly sensitive to hyperparameters, and outputs were often inconsistent across runs.

SinGAN, while promising in low-data regimes, exhibited highly unstable training behavior. Finding a configuration that produced usable results proved challenging, as the model was extremely sensitive to hyperparameters. However, the best configuration (Comb_Exp4) succeeded in generating visually realistic textures. A persistent limitation was the presence of central distortions, which likely stem from the model's original design for natural image synthesis—less suited to the structured and grayscale nature of PBF images.

Stable Diffusion outperformed all other models in both quantitative metrics and visual inspection, consistently producing images that were remarkably similar to real ones. Even the pretrained model, without any fine-tuning, yielded high-quality results. Its ability to incorporate domain-specific prompts and further adapt via DreamBooth LoRA fine-tuning enabled the generation of highly realistic and structurally coherent images. Despite being computationally intensive, diffusion models proved to be the most robust and flexible solution for synthetic data generation in this context.

The promising results of this study suggest several directions for future research. One key opportunity lies in using synthetic data for defect localization and segmentation, enabling models to not only detect but also precisely delineate defect regions. Expanding the current binary classification setup to multi-class defect classification would allow for the identification of different defect types, better reflecting real-world scenarios. Furthermore, advanced prompt engineering—through the use of richer and more descriptive inputs—could enhance both the diversity and controllability of the generated images, leading to more effective and targeted training data.

In conclusion, generative models—particularly diffusion-based approaches—offer a powerful solution for overcoming data limitations in Additive Manufacturing.

REFERENCES

- [1] M. Taghian, M. H. Mosallanejad, E. Lannunziata, G. Del Greco, L. Iuliano, and A. Saboori, "Laser powder bed fusion of metallic components: Latest progress in productivity, quality, and cost perspectives," *Journal of Materials Research and Technology*, vol. 27, pp. 6484–6500, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2238785423028235>
- [2] S. Sun, M. Brandt, and M. Easton, "2 - powder bed fusion processes: An overview," in *Laser Additive Manufacturing*, ser. Woodhead Publishing Series in Electronic and Optical Materials, M. Brandt, Ed. Woodhead Publishing, 2017, pp. 55–77. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/B9780081004333000026>
- [3] Y. Chen, Z. Yan, and Y. Zhu, "A comprehensive survey for generative data augmentation," *Neurocomputing*, vol. 600, p. 128167, 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S092523122400938X>
- [4] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," 2022. [Online]. Available: <https://arxiv.org/abs/1312.6114>
- [5] K. Sohn, H. Lee, and X. Yan, "Learning structured output representation using deep conditional generative models," in *Advances in Neural Information Processing Systems*, C. Cortes, N. Lawrence, D. Lee, M. Sugiyama, and R. Garnett, Eds., vol. 28. Curran Associates, Inc., 2015. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2015/file/8d55a249e6baa5c06772297520da2051-Paper.pdf
- [6] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," 2014. [Online]. Available: <https://arxiv.org/abs/1406.2661>
- [7] T. R. Shaham, T. Dekel, and T. Michaeli, "Singan: Learning a generative model from a single natural image," 2019. [Online]. Available: <https://arxiv.org/abs/1905.01164>
- [8] L. Yang, Z. Zhang, Y. Song, S. Hong, R. Xu, Y. Zhao, W. Zhang, B. Cui, and M.-H. Yang, "Diffusion models: A comprehensive survey of methods and applications," 2024. [Online]. Available: <https://arxiv.org/abs/2209.00796>
- [9] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," 2016. [Online]. Available: <https://arxiv.org/abs/1511.06434>
- [10] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," 2022. [Online]. Available: <https://arxiv.org/abs/2112.10752>
- [11] S. Vivekananthan, "Comparative analysis of generative models: Enhancing image synthesis with vaes, gans, and stable diffusion," 2024. [Online]. Available: <https://arxiv.org/abs/2408.08751>
- [12] T. Xia, "Penalty gradient normalization for generative adversarial networks," 2023. [Online]. Available: <https://arxiv.org/abs/2306.13576>
- [13] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, and K. Aberman, "Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation," 2023. [Online]. Available: <https://arxiv.org/abs/2208.12242>
- [14] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," 2021. [Online]. Available: <https://arxiv.org/abs/2106.09685>
- [15] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," 2018. [Online]. Available: <https://arxiv.org/abs/1801.03924>

APPENDIX A

VAES EXPERIMENTAL CONFIGURATIONS

This appendix details the configuration parameters used in the experiments conducted with the VAE architecture.

Exp	Data_Dir	n_epochs	Img_Size	Latent_Dim	lr	KLD_weight
1	Defects	100	512	128	0.001	1
2	Defects	100	512	256	0.001	1
3	Defects	500	512	256	0.001	1
4	Defects	300	512	128	0.0001	1
5	Defects	300	512	128	0.001	1
6	Defects	300	800	128	0.001	1
7	Defects	600	512	256	0.001	1
8	Defects	200	512	128	0.001	0.5
9	Defects	500	512	128	0.001	0.5
10	Defects	500	512	128	0.001	0.7
11	Defects	700	512	128	0.001	0.5
12	NoDefects	100	512	128	0.001	1
13	NoDefects	100	512	256	0.001	1
14	NoDefects	500	512	256	0.001	1
15	NoDefects	300	512	128	0.0001	1
16	NoDefects	300	512	128	0.001	1
17	NoDefects	300	800	128	0.001	1
18	NoDefects	600	512	256	0.001	1
19	NoDefects	200	512	128	0.001	0.5
20	NoDefects	500	512	128	0.001	0.5
21	NoDefects	500	512	128	0.001	0.7
22	NoDefects	700	512	128	0.001	0.5

Table VI: Summary of experimental configurations for VAE training runs.

The fused experiments are: 1 and 12 (Fusion_1_12), 2 and 13 (Fusion_2_13), 3 and 14 (Fusion_3_14), 4 and 15 (Fusion_4_15), 5 and 16 (Fusion_5_16), 6 and 17 (Fusion_6_17), 7 and 18 (Fusion_7_18), 8 and 19 (Fusion_8_19), 9 and 20 (Fusion_9_20), 10 and 21 (Fusion_10_21), and 11 and 22 (Fusion_11_22).

APPENDIX B

VAES QUALITATIVE RESULTS

This section provides sample synthetic images from each experiment listed in Table VI. Each row displays a fusion experiment, where the first two images belong to the *Defects* class, while the last two correspond to the *NoDefects* class.

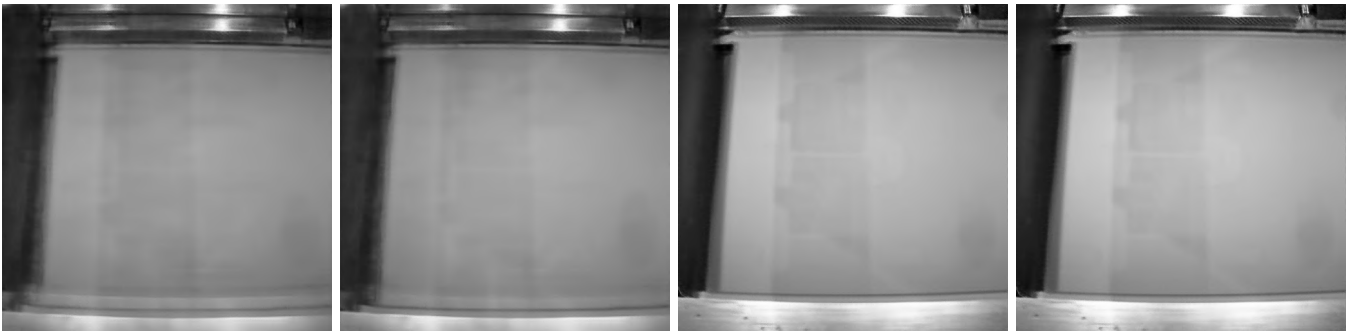


Figure 8: Qualitative results of Fusion_1_12. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

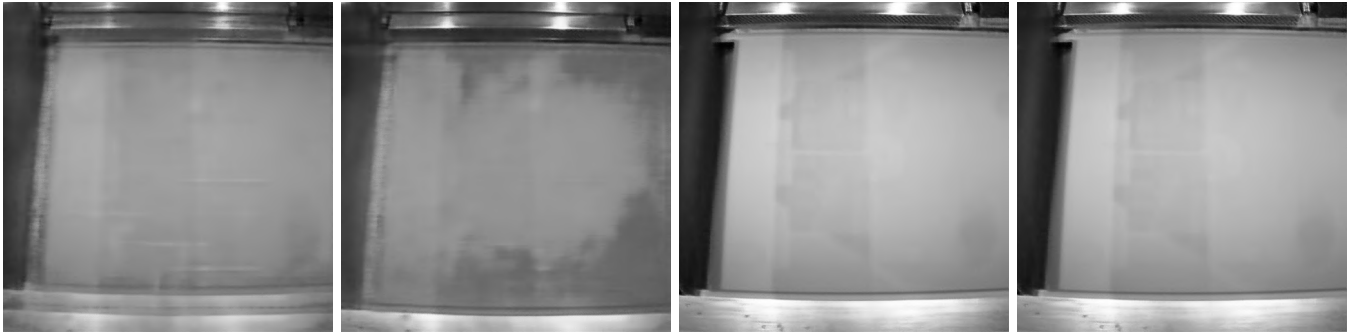


Figure 9: Qualitative results of Fusion_2_13. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

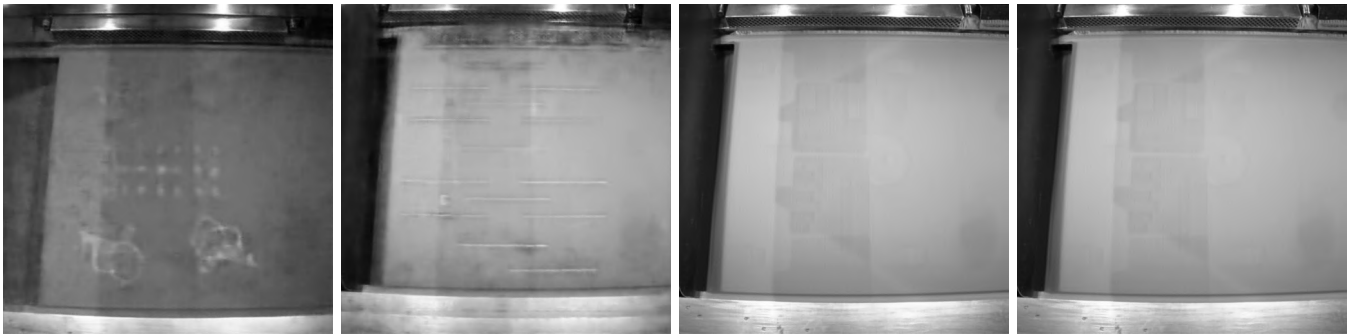


Figure 10: Qualitative results of Fusion_3_14. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.



Figure 11: Qualitative results of Fusion_4_15. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.



Figure 12: Qualitative results of Fusion_5_16. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

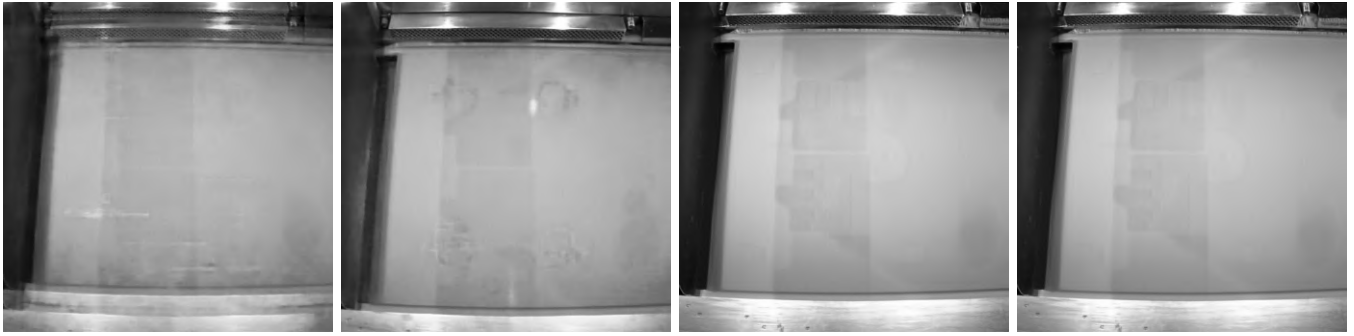


Figure 13: Qualitative results of Fusion_6_17. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

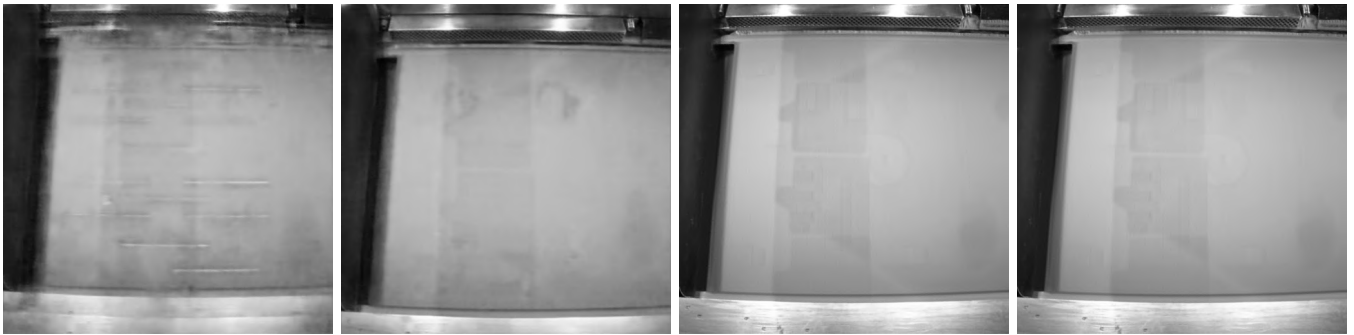


Figure 14: Qualitative results of Fusion_7_18. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

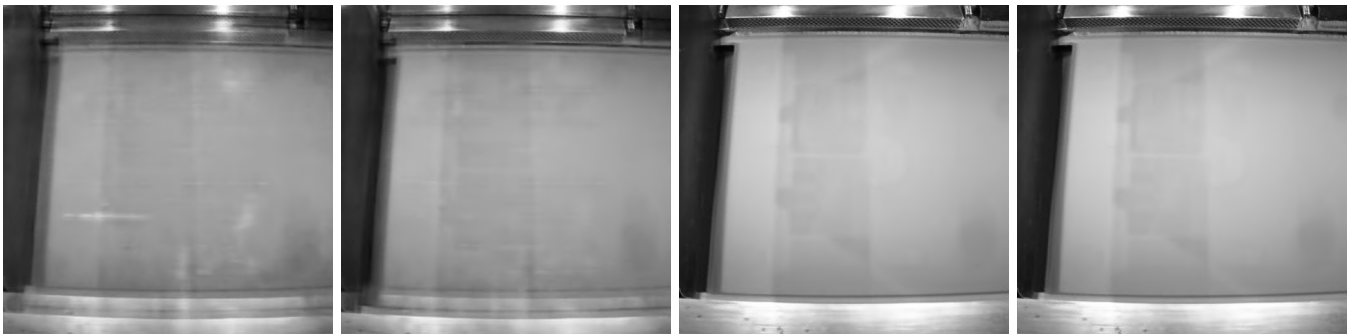


Figure 15: Qualitative results of Fusion_8_19. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

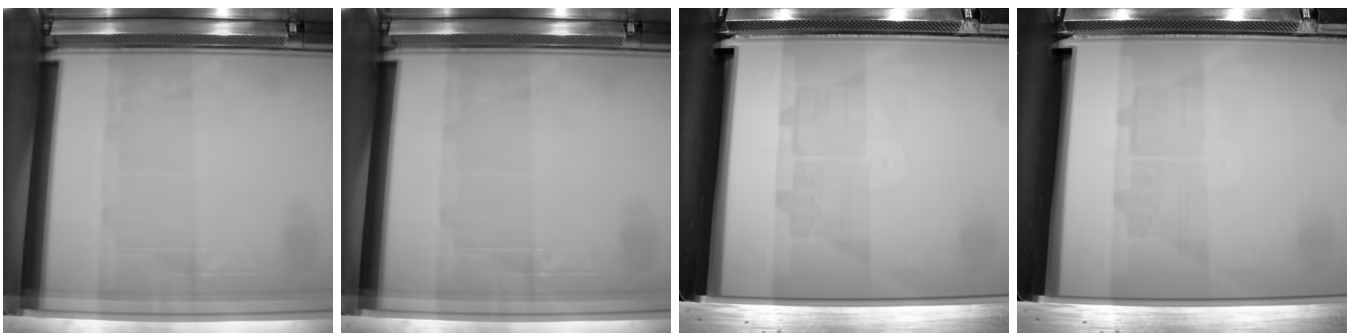


Figure 16: Qualitative results of Fusion_9_20. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

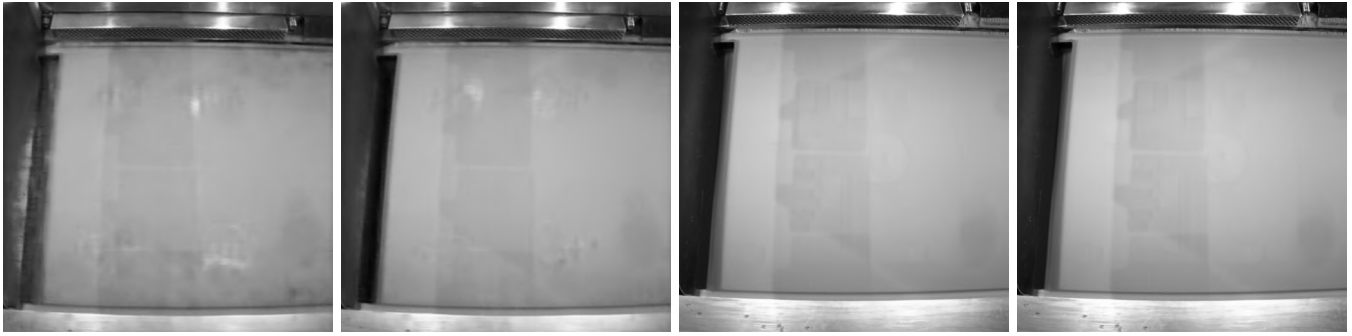


Figure 17: Qualitative results of Fusion_10_21. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

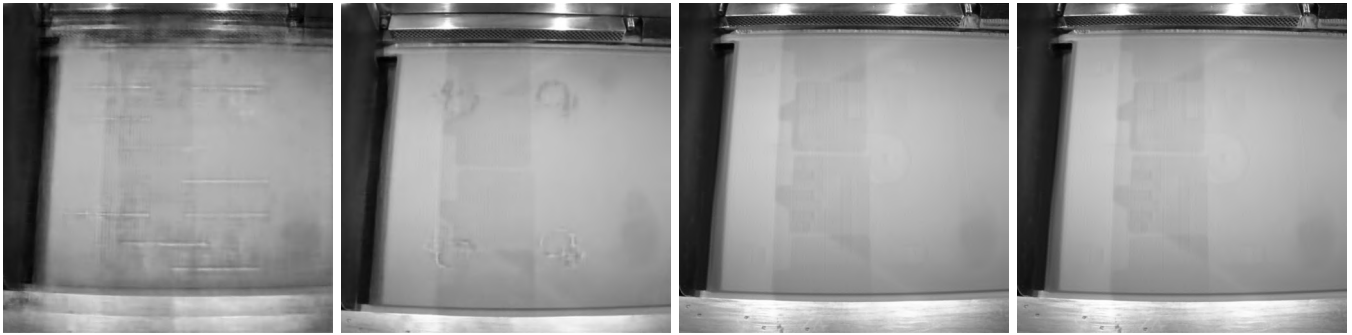


Figure 18: Qualitative results of Fusion_11_22. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

APPENDIX C

CVAES EXPERIMENTAL CONFIGURATIONS

This appendix details the configuration parameters used in the experiments conducted with the CVAE architecture.

Exp	n_epochs	Img_Size	Latent_Dim	lr	KLD_weight
1	100	512	128	0.001	1
2	100	512	256	0.001	1
3	500	512	256	0.001	1
4	300	512	128	0.0001	1
5	300	512	128	0.001	1
6	300	800	128	0.001	1
7	600	512	256	0.001	1
8	200	512	128	0.001	0.5
9	500	512	128	0.001	0.5
10	500	512	128	0.001	0.7
11	700	512	128	0.001	0.5

Table VII: Summary of experimental configurations for CVAE training runs.

APPENDIX D

CVAES QUALITATIVE RESULTS

This section provides sample synthetic images from each experiment listed in Table VII. Each row displays four images: the first two belong to the *Defects* class, while the last two correspond to the *NoDefects* class.

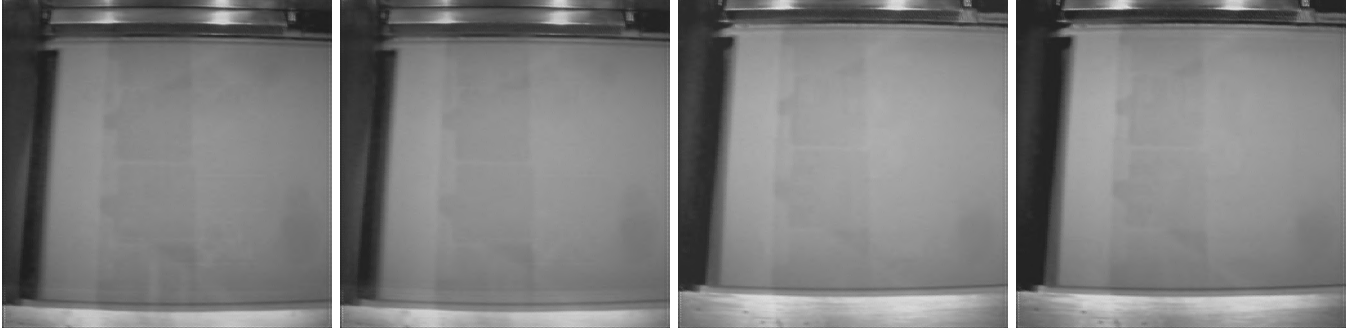


Figure 19: Qualitative results of Exp1. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

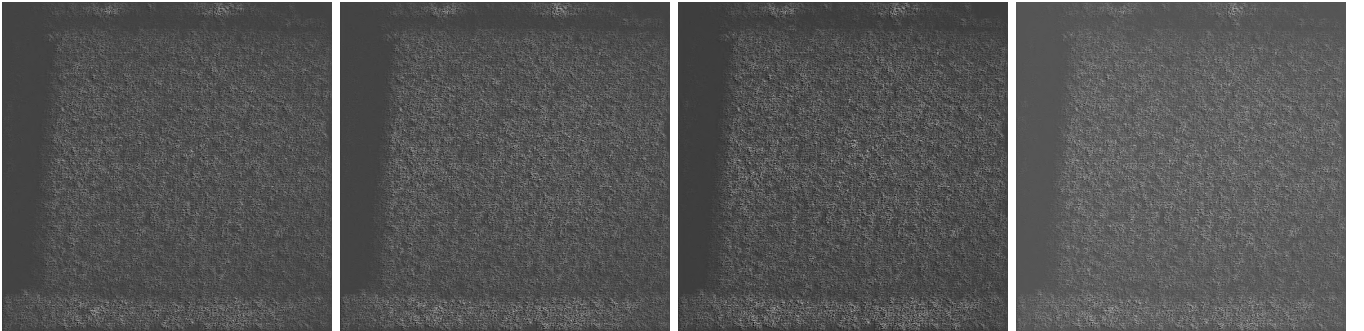


Figure 20: Qualitative results of Exp2. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

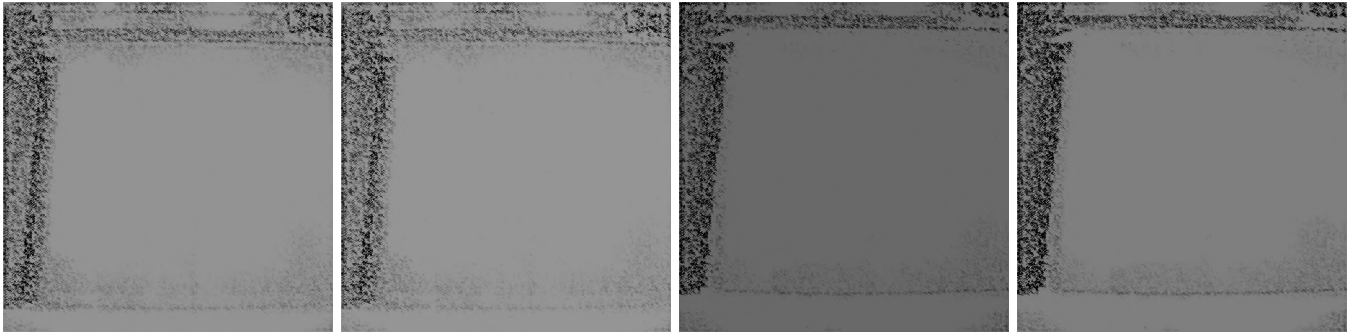


Figure 21: Qualitative results of Exp3. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

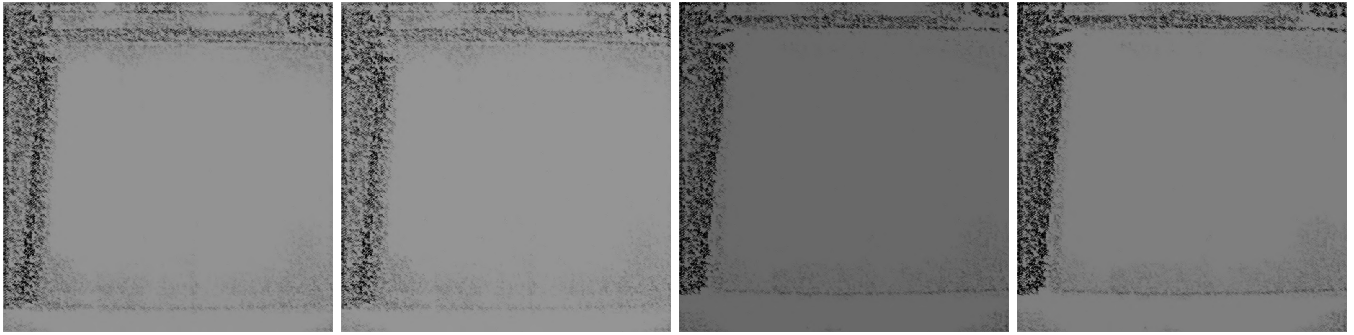


Figure 22: Qualitative results of Exp4. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

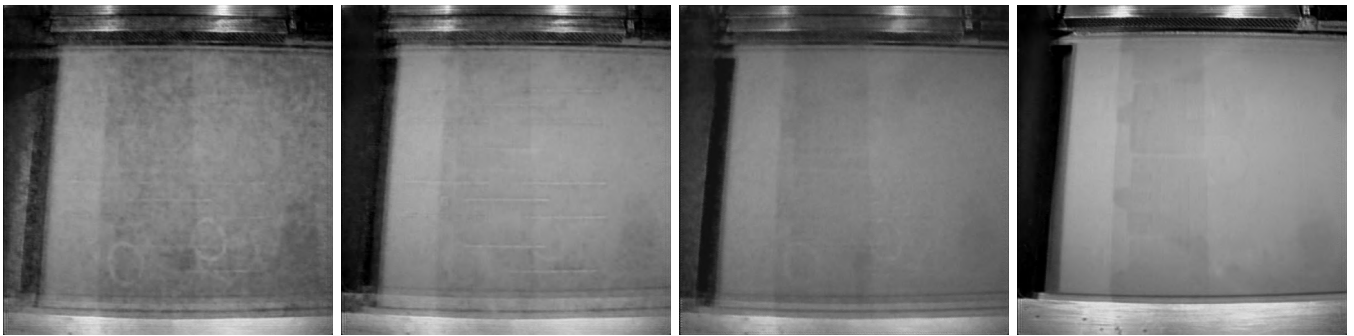


Figure 23: Qualitative results of Exp5. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

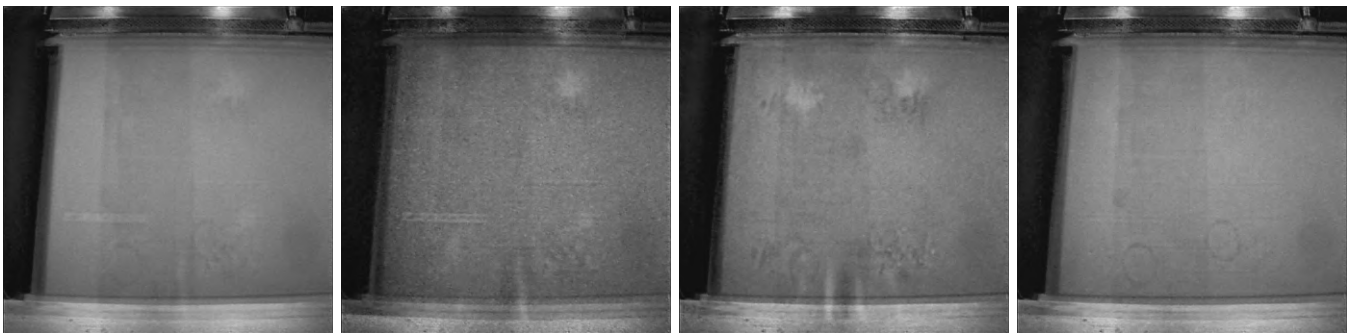


Figure 24: Qualitative results of Exp6. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

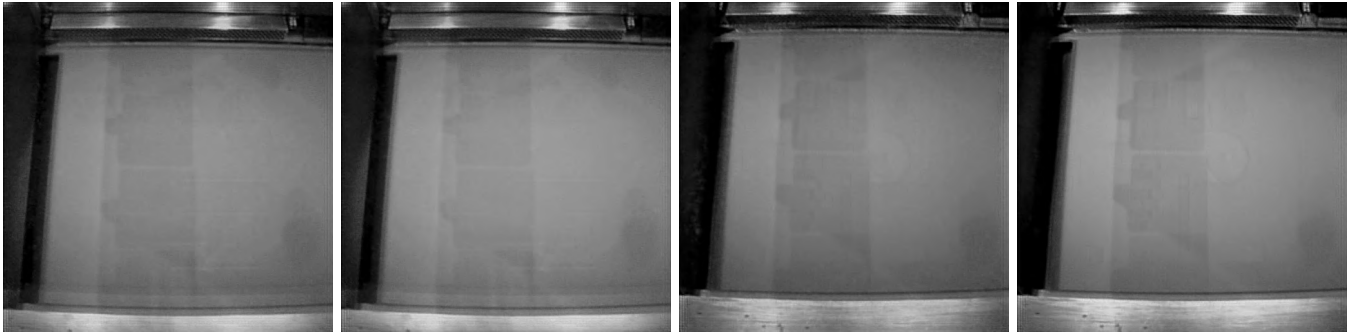


Figure 25: Qualitative results of Exp7. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

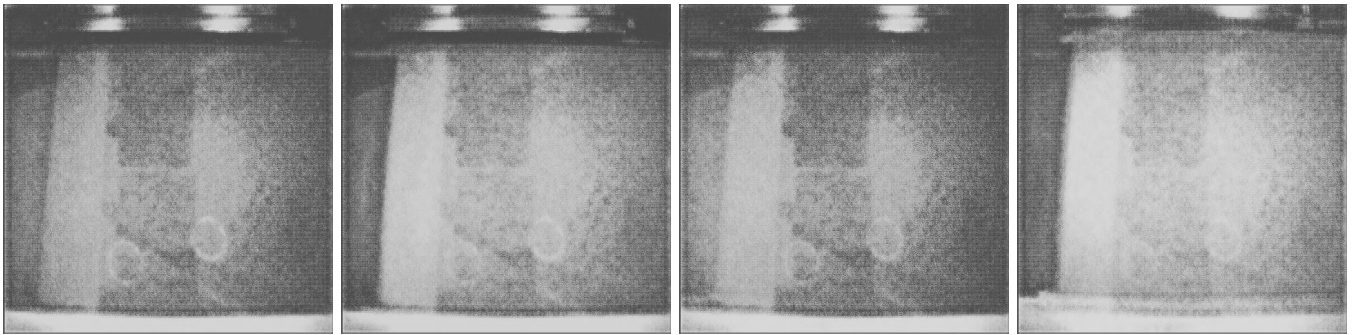


Figure 26: Qualitative results of Exp8. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

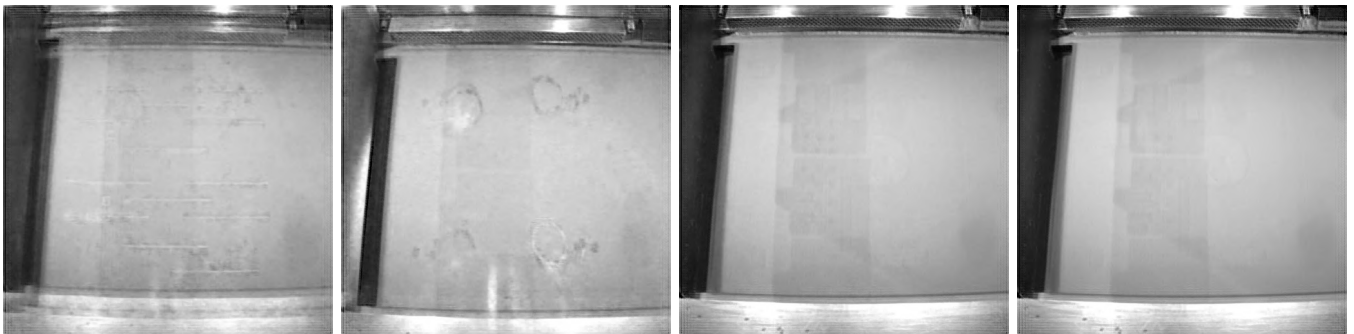


Figure 27: Qualitative results of Exp9. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

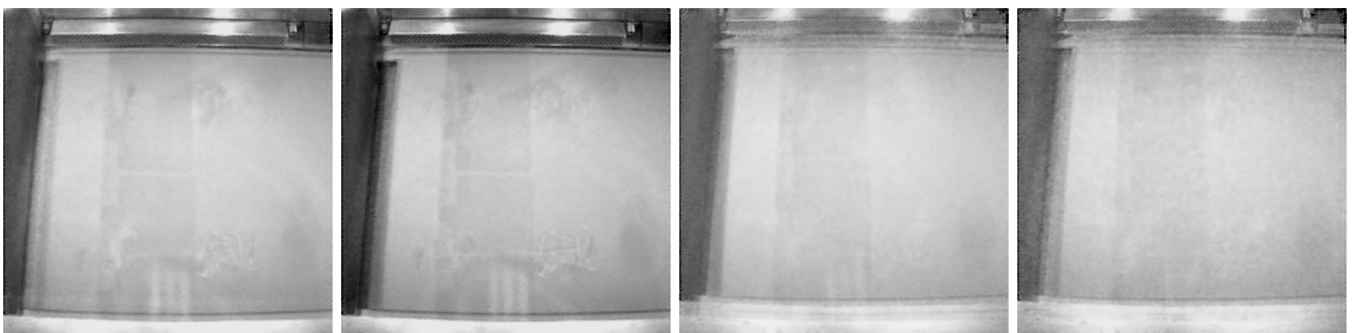


Figure 28: Qualitative results of Exp10. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

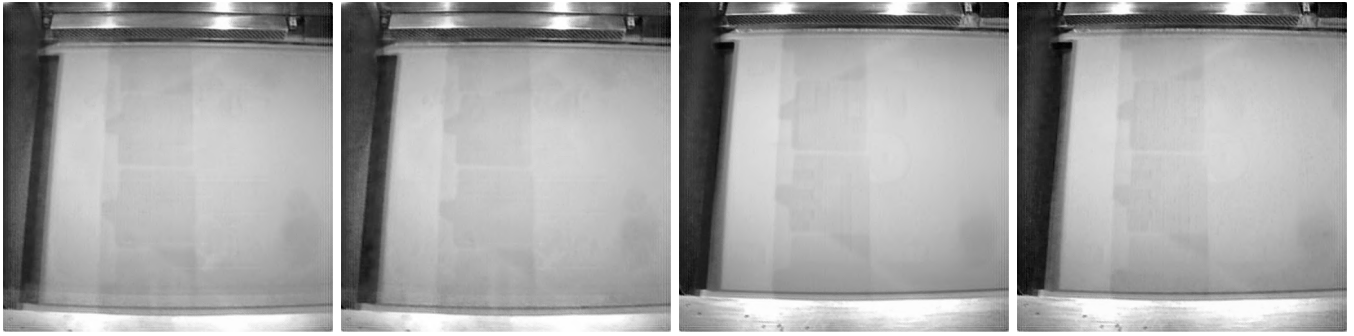


Figure 29: Qualitative results of Exp11. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

APPENDIX E

GANs EXPERIMENTAL CONFIGURATIONS

This appendix details the configuration parameters used in the experiments conducted with the GAN architecture.

Exp	Data_Dir	n_epochs	Img_Size	Latent_Dim	lr_g	lr_d	R1_Reg
1	Defects	1500	512	128	0.0002	0.0002	✗
2	Defects	600	800	128	0.0002	0.0002	✗
3	Defects	600	512	256	0.0002	0.0002	✗
4	Defects	600	800	128	0.0002	0.00002	✗
5	Defects	600	800	128	0.00002	0.0002	✗
6	NoDefects	1500	512	128	0.0002	0.0002	✗
7	Defects	600	800	128	0.0002	0.0002	✓
8	Defects	800	1024	128	0.0002	0.0002	✗
9	NoDefects	800	1024	128	0.0002	0.0002	✗
10	Defects	800	1024	256	0.0002	0.0002	✗
11	NoDefects	800	1024	256	0.0002	0.0002	✗
12	Defects	1000	800	256	0.0002	0.0002	✗
13	NoDefects	1000	800	256	0.0002	0.0002	✗
14	NoDefects	600	800	128	0.0002	0.0002	✗
15	Defects	1200	1024	128	0.0002	0.0002	✗
16	Defects	1200	1024	128	0.0002	0.0002	✓
17	Defects	2000	800	256	0.0002	0.0002	✗
18	Defects	2000	800	256	0.0002	0.0002	✓

Table VIII: Summary of experimental configurations for DCGAN training runs.

The complementary experiments are: 1 and 6 (Fusion_1_6), 2 and 14 (Fusion_2_14), 8 and 9 (Fusion_9_8), 15 and 9 ((Fusion_9_15)), 16 and 9 ((Fusion_9_16)), 10 and 11 (Fusion_10_11, both failed due to cuda out of memory), 12 and 13 (Fusion_13_12), 17 and 13 (Fusion_13_17), and 18 and 13 (Fusion_13_18).

APPENDIX F

GAN QUALITATIVE RESULTS

This section provides sample synthetic images from each experiment listed in Table VIII.

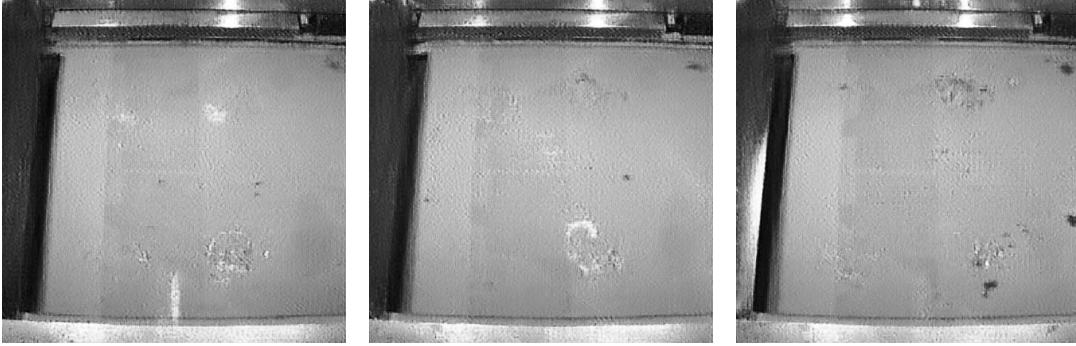


Figure 30: Qualitative results of Exp1 (*Defects*)

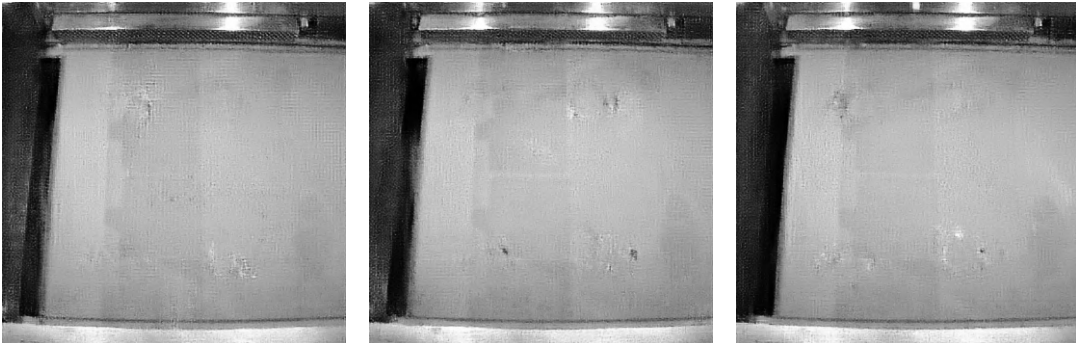


Figure 31: Qualitative results of Exp2 (*Defects*)

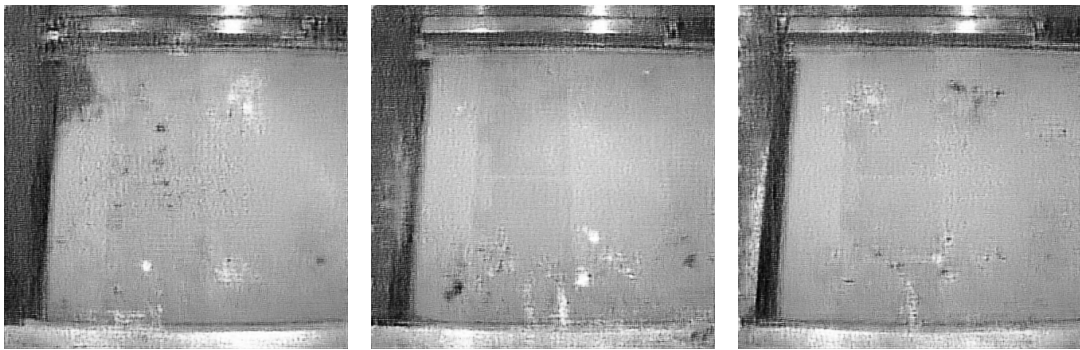


Figure 32: Qualitative results of Exp3 (*Defects*)

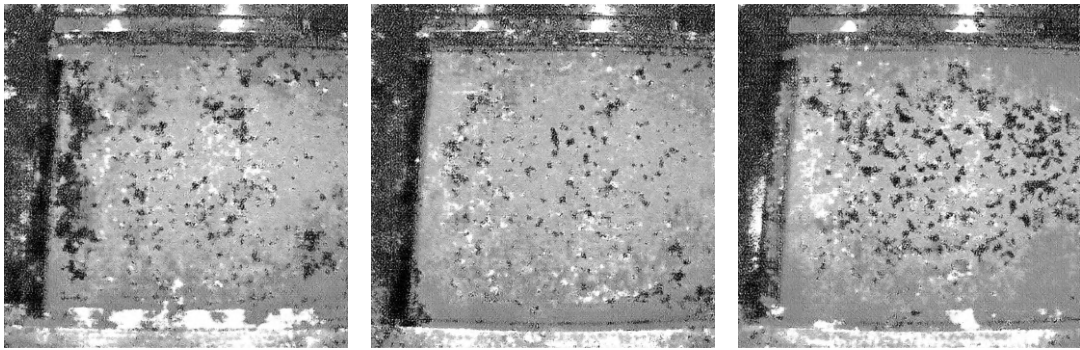


Figure 33: Qualitative results of Exp4 (*Defects*)

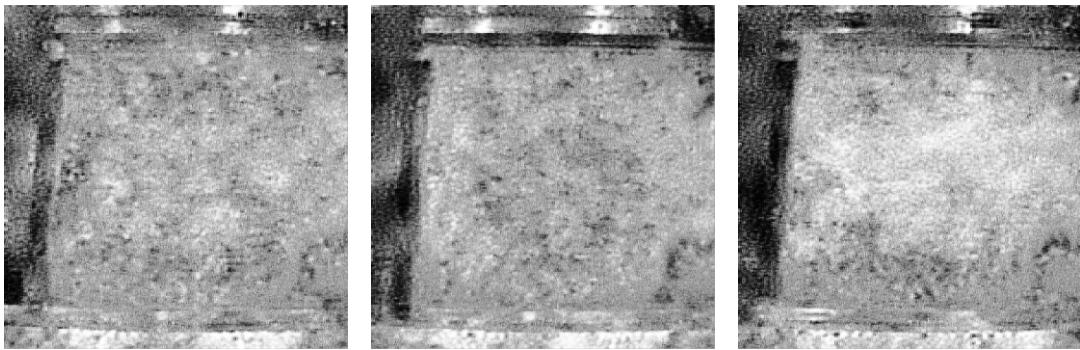


Figure 34: Qualitative results of Exp5 (*Defects*)

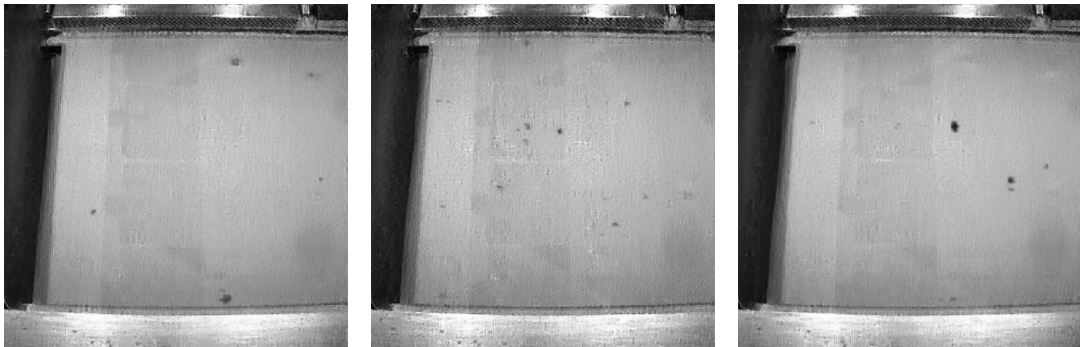


Figure 35: Qualitative results of Exp6 (*NoDefects*)

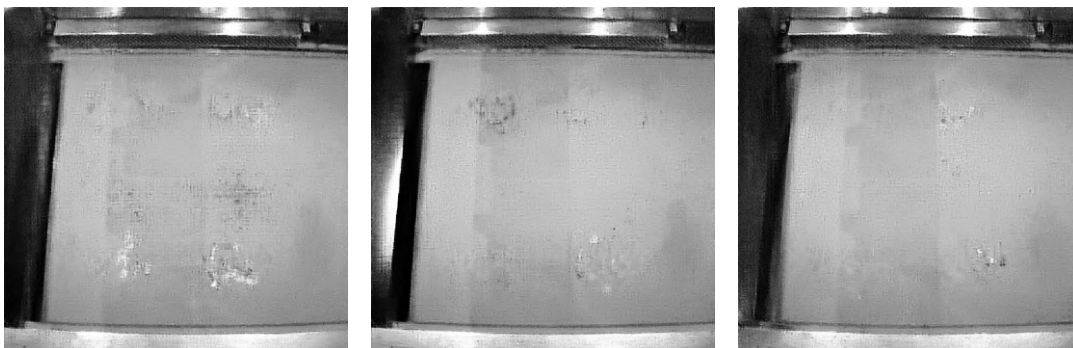


Figure 36: Qualitative results of Exp7 (*Defects*)

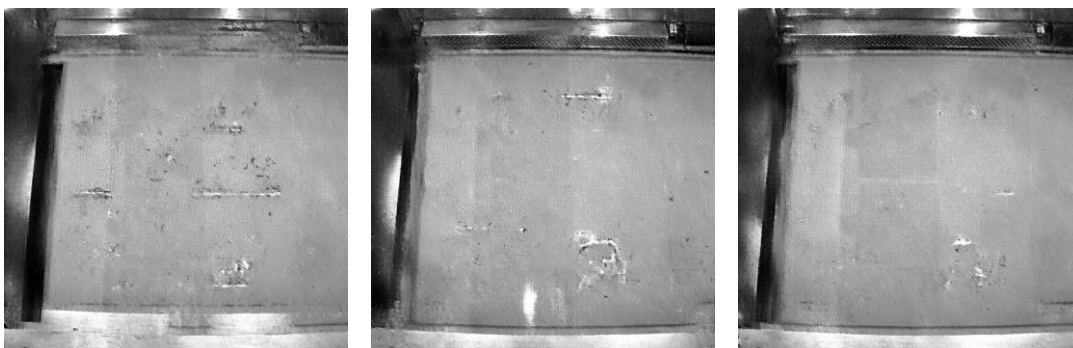


Figure 37: Qualitative results of Exp8 (*Defects*)

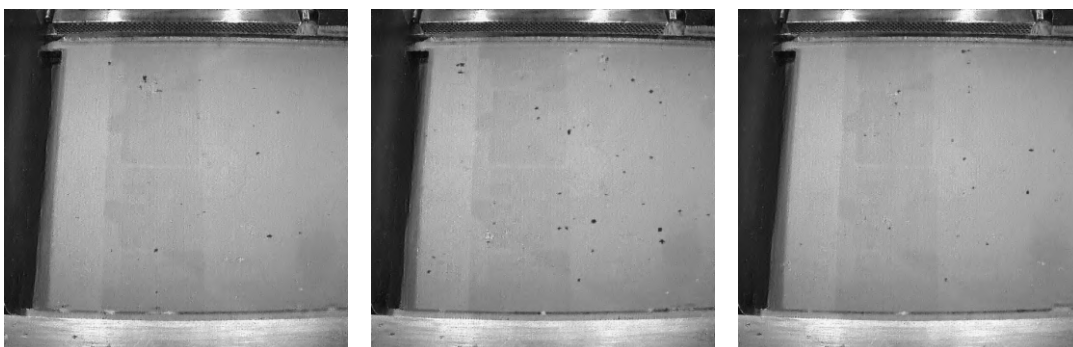


Figure 38: Qualitative results of Exp9 (*NoDefects*)

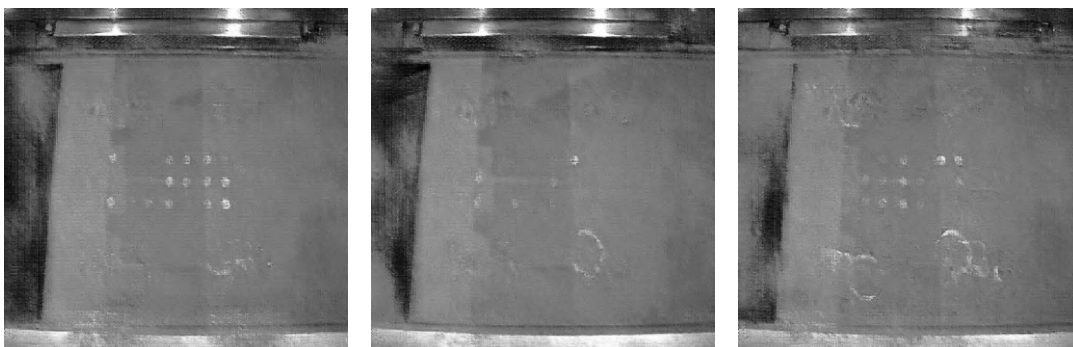


Figure 39: Qualitative results of Exp12 (*Defects*)

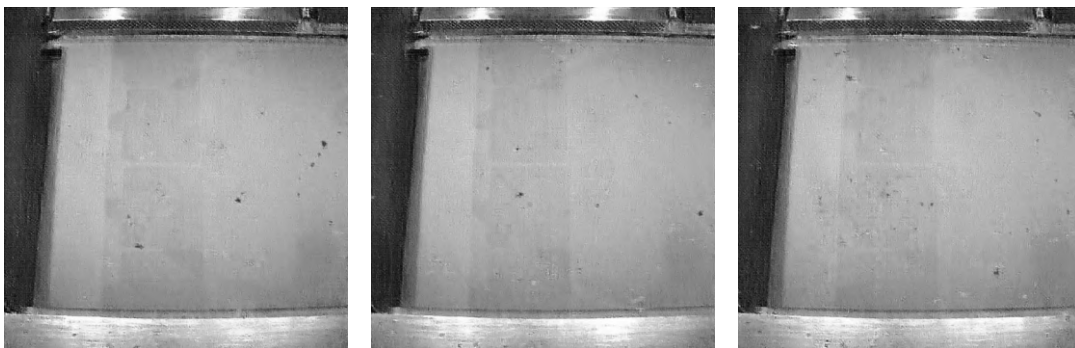


Figure 40: Qualitative results of Exp13 (*NoDefects*)

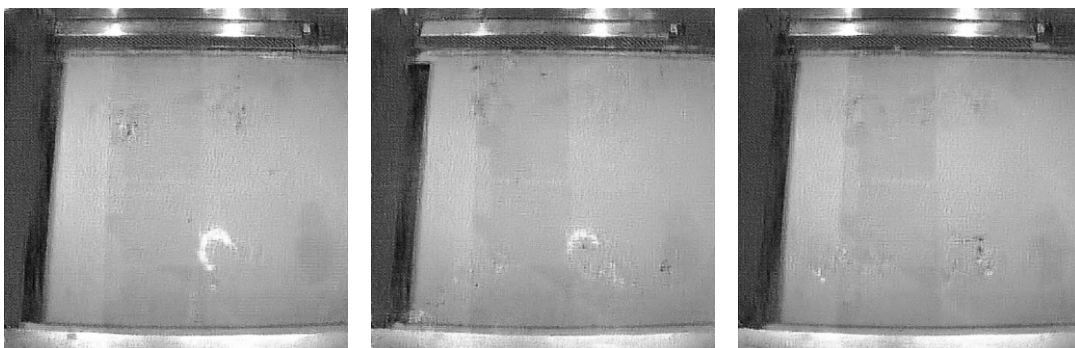


Figure 41: Qualitative results of Exp14 (*NoDefects*)

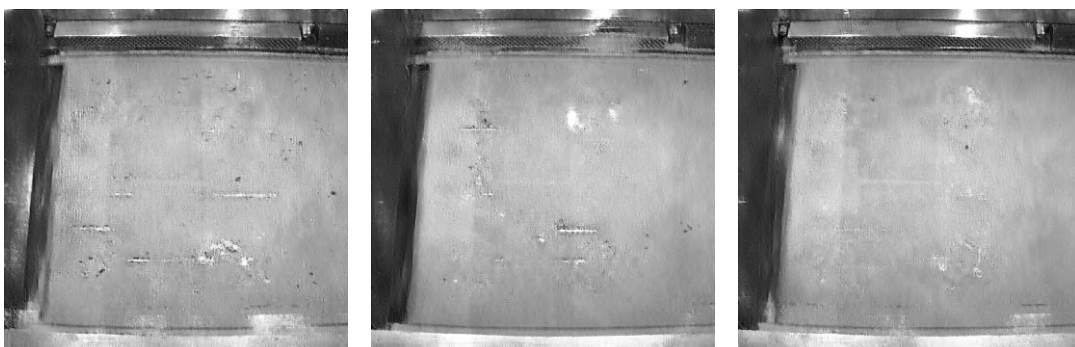


Figure 42: Qualitative results of Exp15 (*Defects*)

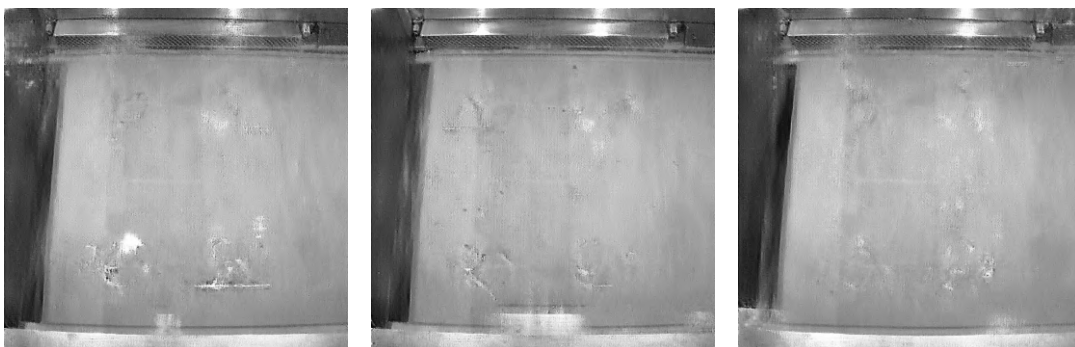


Figure 43: Qualitative results of Exp16 (*Defects*)

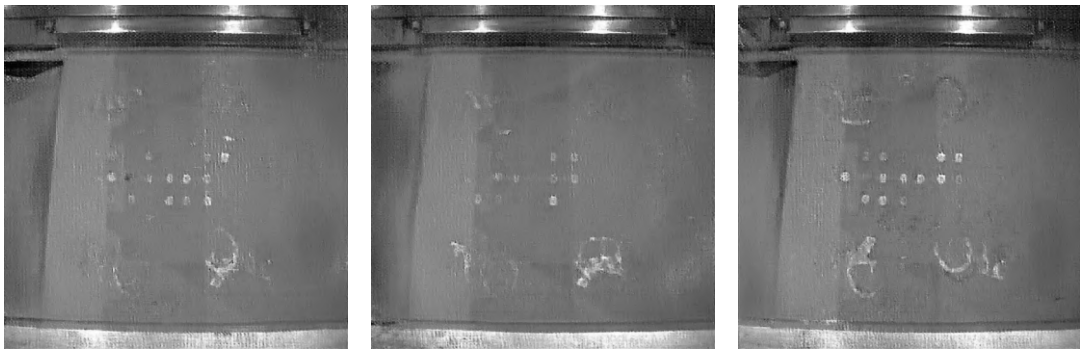


Figure 44: Qualitative results of Exp17 (*Defects*)

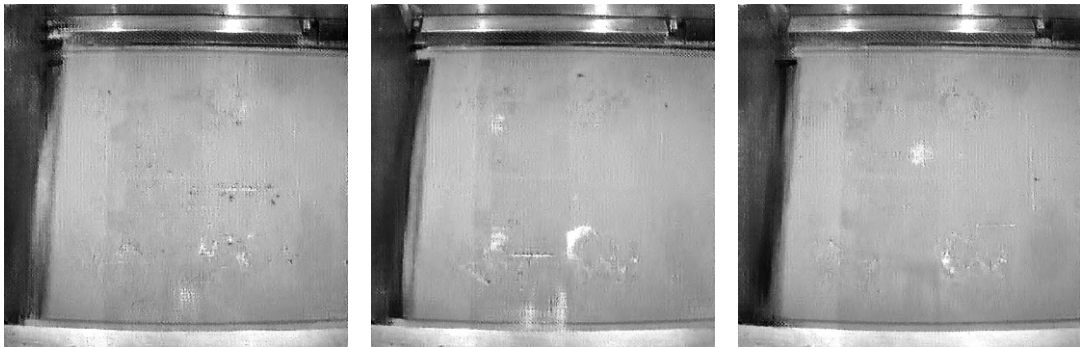


Figure 45: Qualitative results of Exp18 (*Defects*)

APPENDIX G

SINGAN EXPERIMENTAL CONFIGURATIONS

This appendix details the configuration parameters used in the experiments conducted with the SinGAN architecture. Two categories of experiments are presented:

- **Single-image experiments:** Experiments conducted on a single image using custom configurations.
- **Combined experiments:** Experiments where the same configuration is used on multiple images.

Table IX reports the parameters for the single-image experiments. Each row corresponds to an experiment conducted on an individual image with a dedicated configuration.

Exp	scale_factor	niter	lr_g	lr_d	G_steps	D_steps
Exp1	0.50	150	0.001	0.0005	7	5
Exp2	0.80	150	0.001	0.0005	7	5
Exp3	0.80	300	0.001	0.0005	7	5
Exp4	0.80	600	0.001	0.0005	7	5
Exp5	0.80	1000	0.0005	0.0005	5	5
Exp6	0.80	2000	0.001	0.0005	7	5
Exp7	0.90	2000	0.001	0.0005	7	5
Exp8	0.90	2000	0.001	0.0001	7	5
Exp9	0.80	2000	0.001	0.0001	7	5
Exp10	0.80	2000	0.0005	0.00001	7	5
Exp11	0.90	2000	0.001	0.0001	7	5
Exp12	0.80	2000	0.001	0.0005	7	5

Table IX: Configuration parameters for single SinGAN experiments.

Table X reports the parameters for the combined experiments. Each row corresponds to an experiment applied across multiple images.

Comb_Exp	scale_factor	niter	lr_g	lr_d	G_steps	D_steps
Comb_Exp1	0.80	1600	0.001	0.0005	7	5
Comb_Exp2	0.80	2000	0.001	0.0001	7	5
Comb_Exp3	0.90	2000	0.001	0.0001	7	5
Comb_Exp4	0.80	2000	0.0005	0.00001	7	5

Table X: Configuration parameters for combined SinGAN experiments.

APPENDIX H

SINGAN QUALITATIVE RESULTS

This appendix provides a visual overview of each single-image experiment listed in Table IX:

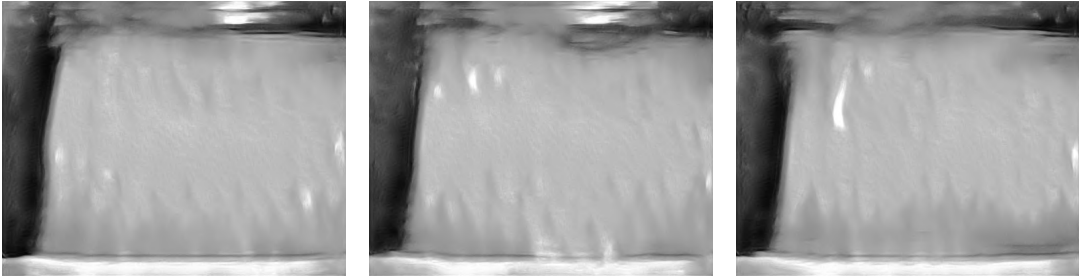


Figure 46: Qualitative results of Exp1 (*Defects*)

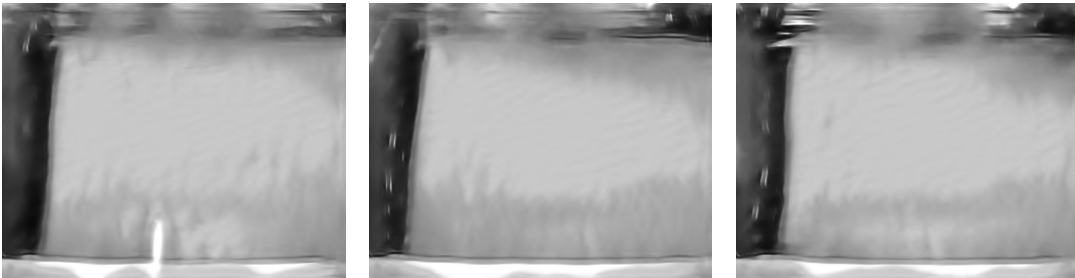


Figure 47: Qualitative results of Exp2 (*Defects*)

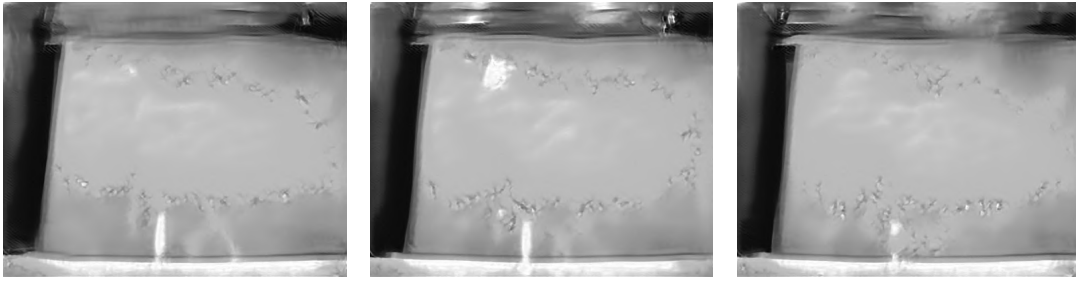


Figure 48: Qualitative results of Exp3 (*Defects*)



Figure 49: Qualitative results of Exp4 (*Defects*)



Figure 50: Qualitative results of Exp5 (*Defects*)



Figure 51: Qualitative results of Exp6 (*Defects*)

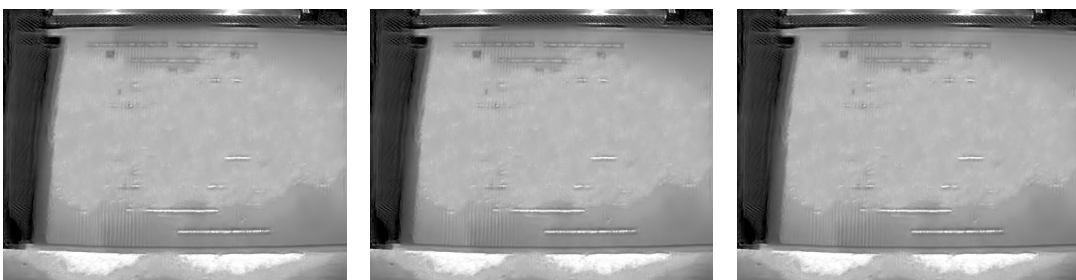


Figure 52: Qualitative results of Exp7 (*Defects*)

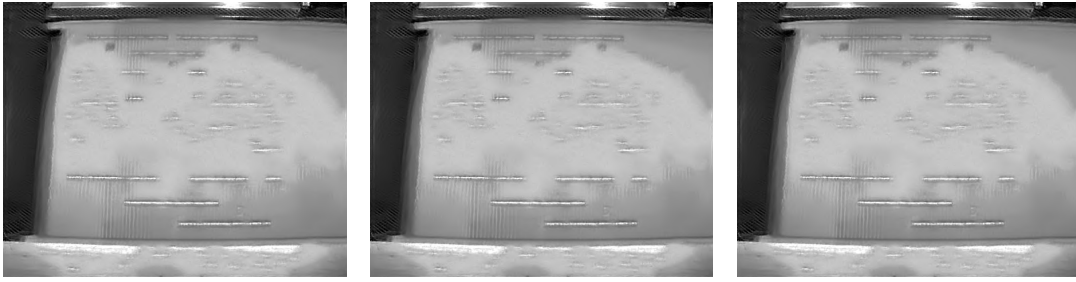


Figure 53: Qualitative results of Exp8 (*Defects*)

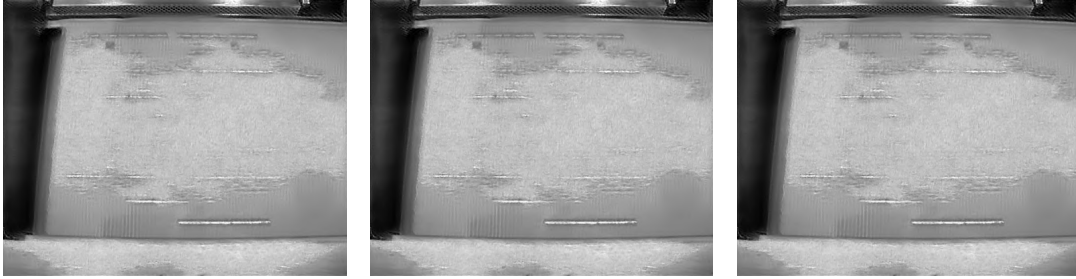


Figure 54: Qualitative results of Exp9 (*Defects*)



Figure 55: Qualitative results of Exp10 (*Defects*)



Figure 56: Qualitative results of Exp11 (*Defects*)

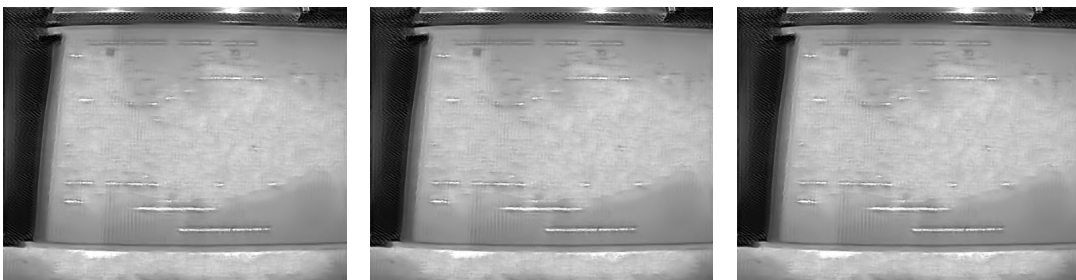


Figure 57: Qualitative results of Exp12 (*Defects*)

This section provides a visual overview of each combined-image experiment listed in Table X, where the first two images (from the left) belong to the *Defects* class, and the remaining two correspond to the *NoDefects* class.

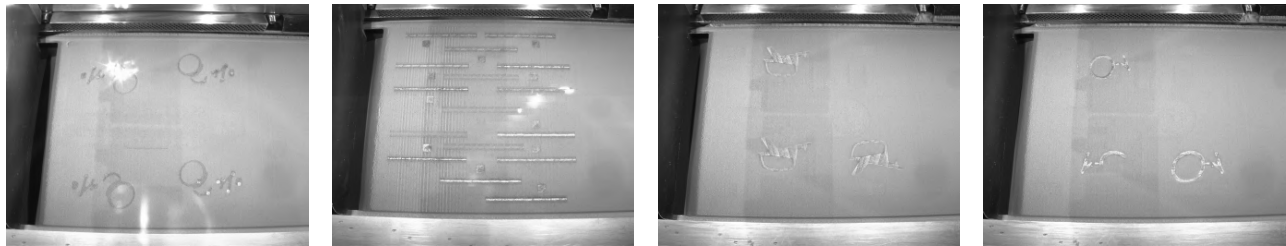


Figure 58: Original Images. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

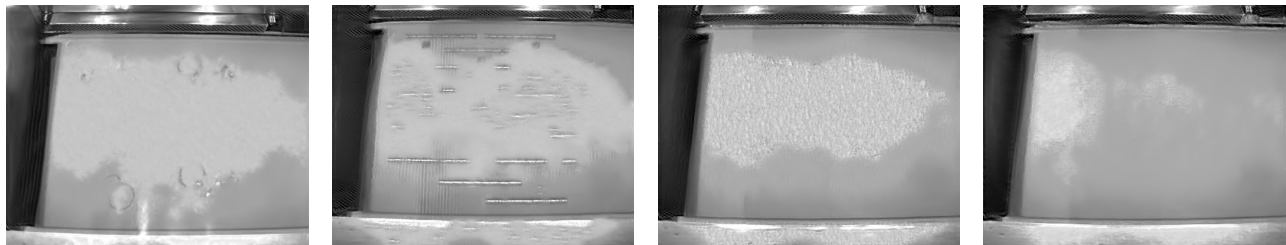


Figure 59: Comb_Exp1. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

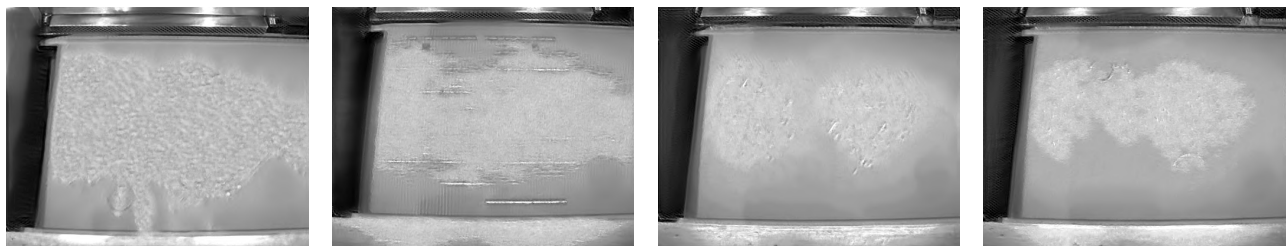


Figure 60: Comb_Exp2. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

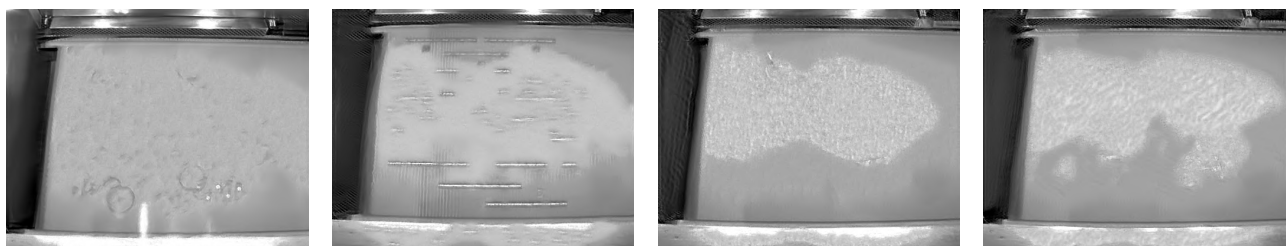


Figure 61: Comb_Exp3. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

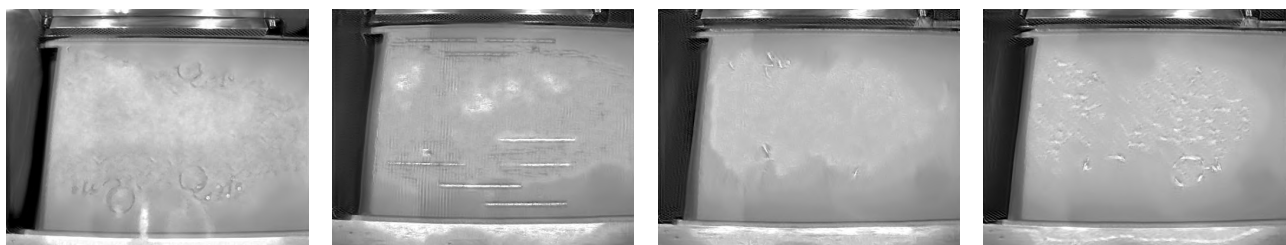


Figure 62: Comb_Exp4. The first two images belong to the *Defects* class, while the last two belong to the *NoDefects* class.

APPENDIX I

STABLEDIFFUSION EXPERIMENTAL CONFIGURATIONS

In this appendix, the configuration parameters used in the experiments conducted with the StableDiffusion models and the parameters used to finetune with Dreambooth-LoRA are listed.

Exp	Steps	Rank	Dropout	Text Enc.	Strength	Guidance	Prompt	Neg. Prompt
Exp1	400	4	–	✗	0.25	5	A	F
Exp2	400	4	–	✗	0.25	8	A	F
Exp3	400	8	–	✗	0.25	8	A	F
Exp4	–	–	–	–	0.15	5	B	F
Exp5	–	–	–	–	0.60	5	B	F
Exp6	400	4	–	✗	0.30	8	C	G
Exp7	400	4	–	✗	0.15	10	D	G
Exp8	800	16	0.1	✓	0.15	8	E	G
Exp9	800	16	0.1	✓	0.25	9	E	G
Exp10	1000	16	0.1	✓	0.25	9	E	G
Exp11	1000	16	0.1	✓	0.25	10	E	G
Exp12	1000	16	0.1	✓	0.15	10	E	G
Exp13	1000	16	0.1	✓	0.15	12	E	G
Exp14	1000	16	0.1	✓	0.20	12	E	G

Table XI: Configuration parameters for StableDiffusion LoRA experiments. Prompt and Negative Prompt refer to the textual variants listed in Table XII.

ID	Prompt Text
A	A high-resolution grayscale technical photo of a powder bed surface in a metal additive manufacturing process, either defect-free or showing typical defects such as spattering, holes, incandescence, or uneven horizontal and vertical lines, with a flat and metallic texture.
B	A grayscale technical photo of a powder bed surface in a metal additive manufacturing process, containing localized defects such as spattering, holes, incandescence, or uneven lines, high-resolution, flat and metallic texture.
C	A high-resolution grayscale technical photo of a metallic powder bed surface in a metal additive manufacturing process, with localized defects randomly displaced across the surface, while keeping the high quality of background and overall scene unchanged. The texture should remain flat and metallic, with no changes to the background, lighting, or general structure.
D	A high-resolution grayscale technical photo of a metallic powder bed surface in a metal additive manufacturing process, with localized objects that should be modified or duplicated randomly across the surface, while keeping the high quality of background and overall scene unchanged. The texture should remain flat and metallic, with no changes to the background, lighting, or general structure.
E	A high-resolution grayscale technical photo of a metallic powder bed surface in a metal additive manufacturing process, with some localized defects (such as spattering, holes, or uneven lines) that have been modified in shape, position, or appearance, or newly introduced in realistic ways. The metallic texture and background should remain unchanged, with consistent lighting and flat structure.

Table XII: Prompt variants (A–E) used across experiments.

ID	Negative Prompt Text
F	artistic, illustrated image
G	artistic, painting, illustration, cartoon, abstract, low quality, blurry, distorted, unrealistic, fantasy, 3d render, color, oversaturated, text, watermark, logo

Table XIII: Negative prompts (F–G) used across experiments.

APPENDIX J
STABLEDIFFUSION QUALITATIVE RESULTS

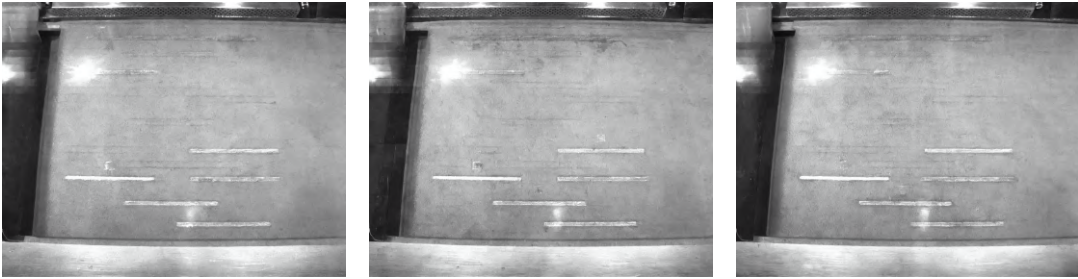


Figure 63: Qualitative results of Exp1 (*Defects*)

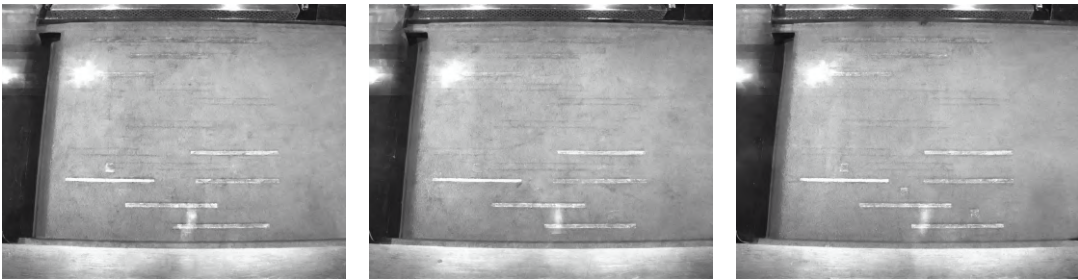


Figure 64: Qualitative results of Exp2 (*Defects*)

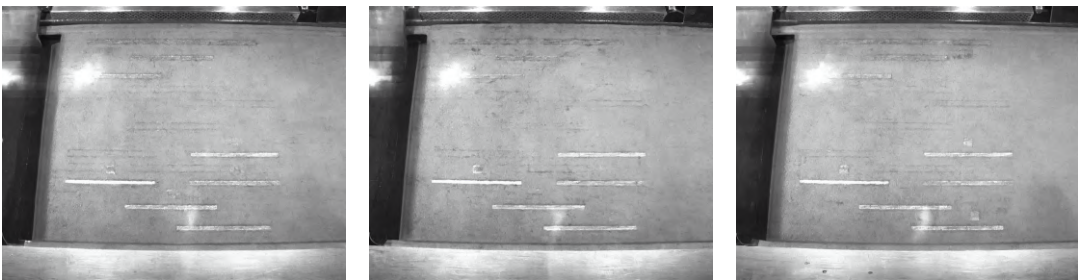


Figure 65: Qualitative results of Exp3 (*Defects*)

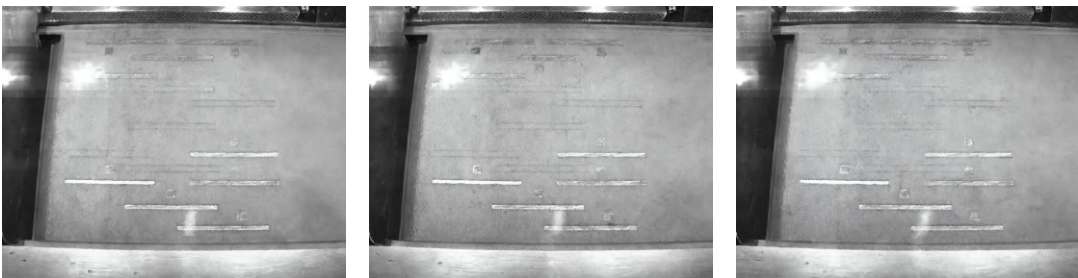


Figure 66: Qualitative results of Exp4 (*Defects*)



Figure 67: Qualitative results of Exp5 (*Defects*)

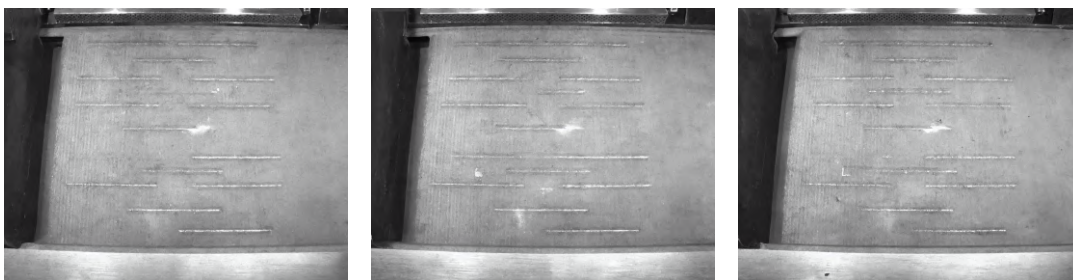


Figure 68: Qualitative results of Exp6 (*Defects*)

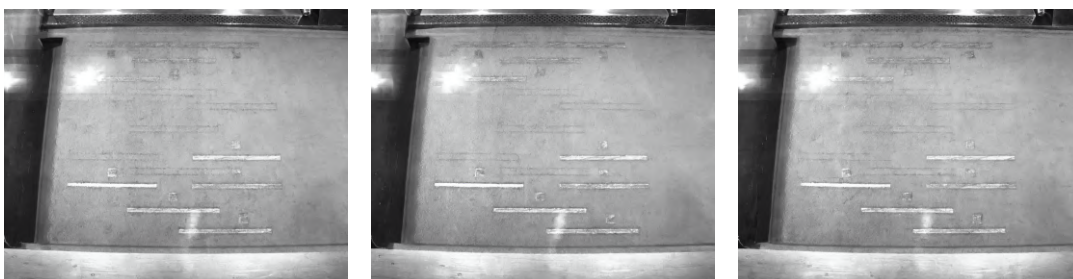


Figure 69: Qualitative results of Exp7 (*Defects*)

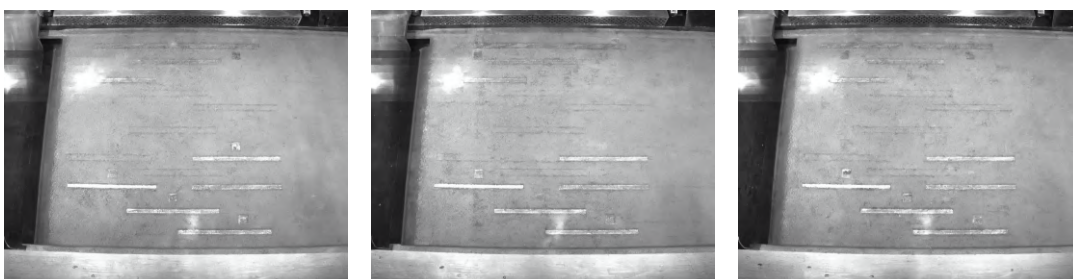


Figure 70: Qualitative results of Exp8 (*Defects*)

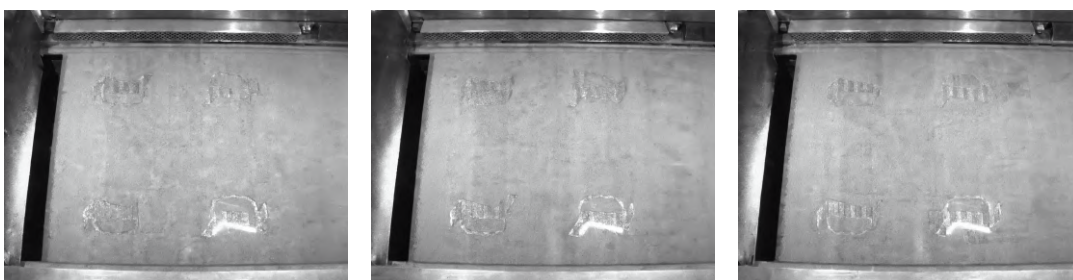


Figure 71: Qualitative results of Exp9 (*Defects*)



Figure 72: Qualitative results of Exp10 (*Defects*)

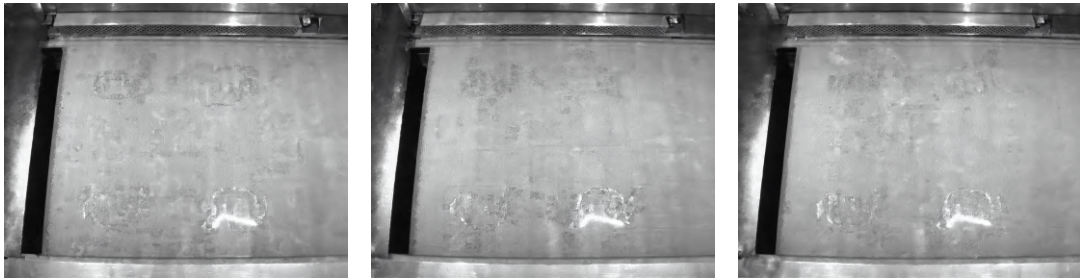


Figure 73: Qualitative results of Exp11 (*Defects*)

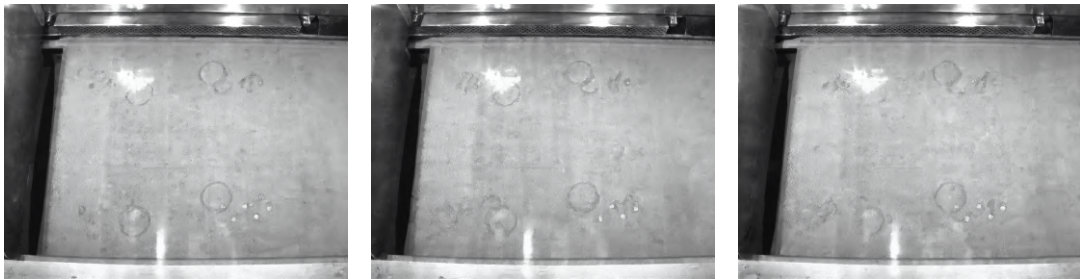


Figure 74: Qualitative results of Exp12 (*Defects*)



Figure 75: Qualitative results of Exp13 (*Defects*)



Figure 76: Qualitative results of Exp14 (*Defects*)